
SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

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ABSTRACT

Although spatial reasoning constitutes a foundational faculty of human cognition, it persists as a formidable impediment for contemporary vision–language models (VLMs). The preponderance of prior work has been predicated upon synthetic or LLM-generated environments encumbered by circumscribed task designs and puzzle-like configurations, and has thereby failed to reproduce the real-world intricacy, visual noise, and heterogeneous spatial relationships that VLMs confront in practice. To redress this deficiency, we introduce **SPATIALAB**, a comprehensive benchmark devised for appraising the spatial reasoning of VLMs within realistic, unconstrained contexts. SPATIALAB comprises 1,400 visual question–answer pairs distributed across six principal categories, namely *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each of which is further decomposed into five subcategories, thereby yielding 30 distinct task types. Insofar as each subcategory encompasses no fewer than 25 questions and each principal category no fewer than 200, the benchmark affords balanced coverage while accommodating both multiple-choice and open-ended evaluation. Systematic experimentation spanning a diverse array of state-of-the-art VLMs, encompassing open- and closed-source systems alongside reasoning-focused and spatially specialized architectures, adduces a pronounced disparity in spatial reasoning competence relative to human performance. Within the multiple-choice regime, InternVL3.5-72B attains 54.93% accuracy as against 87.57% for humans; within the open-ended regime, every model incurs a degradation of approximately 10–25%, with GPT-5-mini registering the highest score at 40.93% as against 64.93% for humans. Collectively, these outcomes accentuate salient limitations in the treatment of complex spatial relationships, depth perception, navigation, and 3D geometry. By furnishing a diverse, real-world evaluation framework, SPATIALAB lays bare the critical challenges and attendant opportunities for advancing the spatial reasoning of VLMs, thereby offering a benchmark calibrated to steer subsequent research toward robust, human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning constitutes a foundational dimension of human cognition, subsuming the comprehension of spatial layouts, the mental imagery and manipulation of objects, and the navigation of physical environments. As such, it underpins an expansive spectrum of applications extending from robotics, autonomous driving, and augmented/virtual reality to geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For artificial agents, that is, autonomous systems that perceive and act within dynamic environments, reliable spatial competence presupposes the interpretation of geometric relations, depth cues, occlusion patterns, and agent-centric perspectives amid scenes that are simultaneously visually noisy and structurally intricate (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Although recent vision–language models have appreciably advanced multimodal representation and language grounding, their spatial judgments remain conspicuously fragile once transposed to realistic environments (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025).

Extant spatial benchmarks, however, are for the most part narrow in scope and simplified in construction. The majority concentrate upon elementary tasks, such as binary spatial relations, coarse-grained

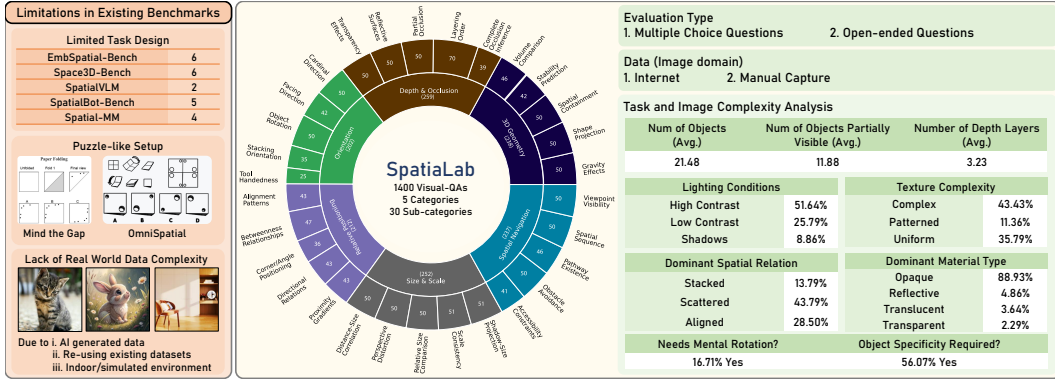


Figure 1: **Overview of SPATIALLAB.** The benchmark redresses the limitations besetting prior datasets (left), introduces 1,400 visual QA pairs spanning 5 categories and 30 subcategories (center), and thereby enables systematic evaluation through both multiple-choice and open-ended tasks. It is distinguished by pronounced task and image complexity, encompassing variation in object counts, layers, lighting, textures, relations, and materials (right).

depth categories, or schematic navigation cues, and situate these tasks within synthetic or otherwise simplified settings that attenuate visual clutter and homogenize object appearances (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024; Qi et al., 2025; Cai et al., 2024). Such controlled configurations diminish the perceptual and inferential burden imposed on the model, thereby engendering an apparent saturation of performance that simultaneously conceals systematic failure under distribution shift. Consequently, critical challenges, among them robust occlusion inference, scale-consistent judgment across divergent viewpoints, and planning under partial observability, remain markedly under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). It follows that models exhibiting strong performance on prevailing benchmarks frequently falter on integrated, multi-step inference, a shortfall that circumscribes their safe deployment in real-world environments wherein spatial errors carry tangible consequences.

A meaningful evaluation must therefore traverse the full complement of core axes constitutive of spatial reasoning, namely relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. In contradistinction to prevailing task-isolated setups, human cognition integrates these dimensions seamlessly even under noise and ambiguity (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). Robust benchmarks ought accordingly to interrogate both selection and generation, to calibrate model performance against human baselines, and to draw upon real-world scenes capable of exposing the failure modes that simulator-derived datasets systematically obscure, principles that collectively motivate the design of our benchmark.

To address the aforementioned needs, we introduce **SPATIALLAB**, a benchmark expressly designed to appraise the spatial reasoning of vision–language models within realistic, unconstrained visual contexts, as delineated in Figure 1. **SPATIALLAB** furnishes a balanced suite of visual question–answer items that spans a broad taxonomy of spatial tasks and accommodates both multiple-choice and open-ended evaluation formats. In its construction, the benchmark foregrounds photographic diversity, compositionally complex spatial-relational queries, and tasks whose resolution necessitates the integration of depth, perspective, and three-dimensional structure. We proceed to evaluate an extensive collection of state-of-the-art VLMs, comprising open- and closed-source models as well as reasoning-focused systems, to juxtapose their behavior against human performance, and to conduct an exhaustive error analysis. Our principal contributions are enumerated as follows:

1. We introduce **SPATIALLAB**, a large-scale benchmark comprising 1,400 visual question–answer pairs that span six principal categories and thirty task types situated in realistic, unconstrained contexts. By virtue of maintaining at least 200 questions per category and 25 per subcategory, it secures balanced coverage while supporting both multiple-choice (SPATIALLAB-MCQ) and open-ended (SPATIALLAB-OPEN) evaluation.

Table 1: A comparative synopsis of spatial reasoning benchmarks.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval		A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.		Type	Depth				
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I	-	-
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I	LXMERT	70.10
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I	XVLM-COCO	60.40
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I	Qwen-VL-Max	49.11
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I	RAG3D-Chat	66.80
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I	GPT-4o	57.83
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I	GPT-4o	59.03
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I	GPT-4 Vision	63.82
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I	LEO	71.90
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I	GPT-4V	68.00
VSI-Bench	Y	8	Indoor	M.E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V	Gemini-1.5 Pro	45.40
OmniSpatial	Y	50	Internet	Internet	M,P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I	o3-2025-04-16	56.33
Mind the Gap	N	6	Mix	AI-Gen	M,P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I	InternVL2.5-26B	48.83
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix	Gemini-2.5 Pro	62.00
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix	InternVL3.5-72B	54.93
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix	GPT-5-mini	40.93

2. We undertake a **comprehensive evaluation** of upwards of 25 state-of-the-art vision-language models, encompassing open- and closed-source, reasoning-oriented, and spatially specialized reasoning models, benchmarked against human baselines, thereby laying bare a consistent and pronounced deficit in spatial reasoning.
3. We furnish **in-depth error analyses and diagnostic studies** that adduce systematic failure modes in depth perception, navigation, occlusion, and 3D geometry, and that thereby yield concrete insights together with actionable directions for advancing robust, human-aligned spatial understanding.
4. We investigate a plurality of strategies for augmenting spatial reasoning in VLMs, namely **supervised fine-tuning (SFT), chain-of-thought (CoT) prompting, CoT with self-reflection, and multi-agent architectures (SPATIOXOLVER) endowed with multi-step reasoning**, and we scrutinize their respective benefits and limitations across multiple-choice and open-ended tasks.

Within the multiple-choice setting, the foremost models attain roughly 50–55% accuracy, whereas human annotators surpass 85%. Under open-ended generation, model performance declines by a further 10–25 percentage points, such that even the strongest systems hover near 40% as against approximately 65% for humans. Our error analysis discloses failures concentrated in occlusion inference, scale-consistency, and multi-step navigational planning, thereby signalling the limits of compositional spatial representation and reasoning. Taken as a whole, our comprehensive study elucidates which interventions confer robust gains, which instead amplify pre-existing biases, and, crucially, the pivotal role that perceptual grounding assumes in generalizable spatial reasoning. By supplying tasks, baseline results, and diagnostic analyses in concert, SPATIALAB establishes a durable foundation for progress and for reproducible comparison in the evaluation of spatial reasoning.

2 PRELIMINARIES: SPATIAL REASONING IN VLMs

Visual-spatial reasoning, understood as the capacity to perceive, represent, and manipulate the spatial relations obtaining among objects, occupies a central position in human cognition and is indispensable to the ability of embodied agents to navigate and act upon the world (Palmer, 1999; Chabris et al., 2006). Formally, a scene may be characterized as a set of objects $O = \{o_1, \dots, o_n\}$ endowed with attributes (e.g., position, orientation, size) referred to a world coordinate system W . Visual-spatial reasoning, so construed, seeks to learn a mapping $f : (I, Q) \mapsto R$, wherein I denotes an image (or image sequence), Q a linguistic query, and R a structured representation of spatial relations $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ that encodes predicates such as *left-of*, *occludes*, or *supports*. For vision-language models (VLMs), the acquisition of f proves exceptionally demanding, inasmuch as it necessitates the integration of noisy visual perception with compositional linguistic abstraction under embodied constraints such as perspective, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022).

Early approaches to spatial reasoning in VLMs have, for the most part, been predicated upon rudimentary spatial relations or narrowly circumscribed task designs (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), and have frequently been situated within synthetic datasets such as CLEVR (Johnson et al., 2016), structured question-answering benchmarks such as GQA (Hudson & Manning, 2019), or puzzle-like simulated configurations such as MGT (Stogiannidis et al., 2025) and

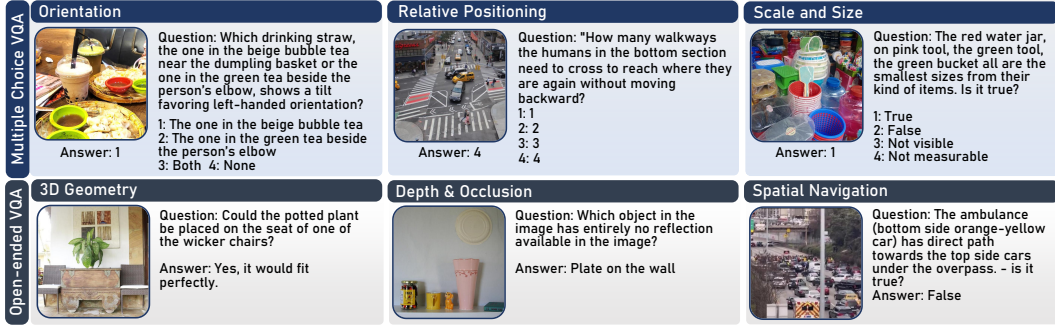


Figure 2: Representative exemplars drawn from the six categories across both open-ended and MCQ tasks.

OmniSpatial (Stogiannidis et al., 2025), as may be discerned in Table 1¹. Although serviceable for controlled evaluation, these settings oversimplify spatial structure, expunge environmental noise, and disregard dynamic factors such as temporal change and viewpoint shift. In order to surmount these limitations, we contend that spatial reasoning must be benchmarked within rich, real-world contexts that faithfully capture scene complexity, temporal variation, and a multiplicity of perspectives. Settings of this kind confront VLMs with precisely the challenges they would encounter across embodied or interactive applications, robotics, AR/VR, and autonomous navigation among them (Mu et al., 2023; Collaboration et al., 2023).

Within SPATIALAB, we organize spatial reasoning into six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each of which is associated with a distinct cognitive faculty documented in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As depicted in Figure 1 and elaborated in Appendix B, every category is further decomposed into five subcategories, yielding a taxonomy of thirty task types that isolate fine-grained reasoning skills such as directional alignment, occlusion inference, and stability prediction. The rationale underpinning this classification scheme is the imperative to encompass the full spectrum of visual-spatial reasoning challenges. By systematically decomposing complex reasoning into measurable and testable units, we render tractable not merely the question of whether models “solve” spatial tasks but also the finer question of which specific dimensions remain fragile or underdeveloped (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

We further adopt two complementary evaluation formats, multiple-choice and open-ended question answering. The former interrogates recognition and fine-grained discrimination among distractors, whereas the latter gauges generative reasoning and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Considered jointly, these formats mirror the reasoning demands of real-world deployment and bring to light divergent strengths and weaknesses in model capabilities. By conjoining a principled taxonomy, diverse real-world imagery, and dual evaluation formats, SPATIALAB constitutes a structured framework for systematic benchmarking. This configuration affords fine-grained model comparison, alignment with human-level performance, and a more penetrating understanding of how robust spatial cognition may be cultivated in artificial agents.

3 SPATIALAB BENCHMARK

Spatial reasoning persists as a recalcitrant challenge for vision-language models (VLMs), most acutely under naturalistic conditions wherein occlusion, perspective distortion, and noisy backgrounds co-occur. In response, we present **SPATIALAB**, a benchmark constructed to evaluate systematically the capacity of VLMs for complex spatial cognition under realistic visual conditions. The benchmark comprises 1,400 visual question-answer pairs marshalled into six major categories, namely *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each of which is further partitioned into five subcategories so as to yield 30

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

distinct task types. Insofar as every subcategory comprises at least 25 QA pairs and each top-level category contributes no fewer than 200, balanced coverage across the taxonomy is thereby assured. **SPATIALAB** accommodates both multiple-choice and open-ended evaluation, and thus enables the assessment of discriminative accuracy in conjunction with generative reasoning and explanatory competence, aligning it with broader multimodal evaluation practice while nonetheless foregrounding spatially specific reasoning. Its pronounced visual diversity, fine-grained annotation, and rigorous quality control render **SPATIALAB** a dependable and scalable testbed; Figure 2 accordingly illustrates representative examples from the six categories, exhibiting both open-ended and multiple-choice formats that convey the richness and difficulty characteristic of real-world spatial reasoning tasks.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The image corpus was curated by means of three complementary strategies: (i) automated large-scale web crawling, (ii) targeted online retrieval of scene-specific images, and (iii) manually captured snapshots gathered across diverse indoor and outdoor environments (Figure 3). This multi-source pipeline guarantees diversity across object categories, environmental conditions, and spatial configurations. In order to capture natural variability explicitly, images were profiled along six meta-dimensions, namely *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). Each such dimension was systematically instantiated in the final corpus, thereby ensuring that the benchmark reflects an expansive spectrum of real-world, in-the-wild scenarios rather than simplified or synthetic configurations. A complexity analysis of the corpus discloses that, on average, an image contains 21.48 objects, of which 11.88 are partially visible, distributed across 3.23 distinct depth layers. The spatial reference chains it elicits (e.g., “object A is left of B, which is behind C”) comprise 2.07 links on average, an indication of the multi-step relational reasoning the benchmark demands. Furthermore, 16.71% of the questions necessitate mental rotation, while 56.07% presuppose object specificity (a particular color, pattern, or other distinguishing detail). Statistics of this kind attest that **SPATIALAB** captures the cluttered, layered, and relationally rich scenes that typify real-world spatial reasoning yet remain underrepresented in synthetic datasets.

Annotation. Annotation unfolded across three successive phases. In the first, annotators underwent targeted training (details of which are provided in Appendix C.2) in order to secure consistency in the interpretation of spatial categories and subcategories. In the second, each image was paired with one or more spatial QA tasks that spanned a range of reasoning types, among them proximity gradients, alignment patterns, occlusion inference, and stability prediction; questions were deliberately constructed so as to balance perceptual grounding against higher-order inference (e.g., predicting stability under gravity constraints). In the third, QA pairs were encoded in both multiple-choice (04 options) and open-ended formats, thereby ensuring comparability across evaluation paradigms. In aggregate, this procedure yields approximately 1,500 spatial visual QA pairs spanning six categories and 30 subcategories.

Review and Quality Control. In order to safeguard reliability, the annotation outputs were subjected to a three-tier review and validation pipeline. At the first stage, QA items were scrutinized for semantic validity and for alignment with the designated subcategory. At the second stage, independent annotators verified the correctness of both the question phrasing and the designated answer, with particular emphasis on the elimination of ambiguity and the enforcement of task clarity. At the third stage, a gold-standard annotation round was conducted so as to establish a set of high-confidence QA pairs for evaluation. All metadata were expunged from the images, and the dataset was assiduously curated to forestall copyright concerns, notwithstanding the possibility of residual minor oversights. This procedure culminated in a final benchmark of 1,400 validated QA items.

4 EXPERIMENTS AND EVALUATION

Models. We stratify the evaluated models into four groups. **Proprietary models** encompass GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** comprise Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-

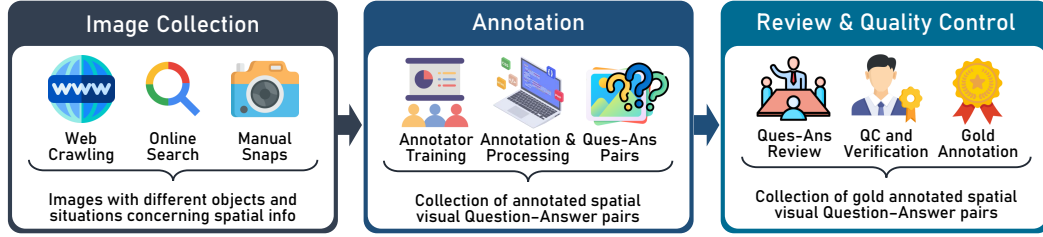


Figure 3: **Data creation pipeline for SPATIALLAB.** Images are amassed by way of web crawling, targeted search, and manually captured snapshots, whereupon they undergo structured annotation into spatial question-answer pairs.

Table 2: Multiple-choice evaluation accuracy (%) (↑) on SPATIALLAB-MCQ, disaggregated by question category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Random Choice	25.00	25.00	25.00	25.00	25.00	25.00	25.00
Proprietary Models							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
Open-Source Models							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
Reasoning Models							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
Spatial Reasoning Models							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
Human Baseline	93.70	74.13	91.58	91.51	88.89	87.76	87.57

MoE (StepFun, 2025). **Reasoning models** include o4-mini (OpenAI, 2025), the Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** comprise SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE we conducted open-ended evaluation exclusively, inasmuch as these models are not instruction-tuned and were trained predominantly on open-ended tasks, a circumstance that precipitated excessive refusals during MCQ evaluation. To facilitate comparison across our results tables, we adopt the following color coding: **proprietary**, **open-source up to 7B**, **larger than 7B**, **proprietary reasoning**, **open-source reasoning**, **spatial specialists**, and **human baseline**.

Evaluation Process and Metrics. We adopt accuracy as the primary metric for both MCQ and open-ended tasks. For **MCQ evaluation**, we employ direct prompting, whereby models return the option number corresponding to their selected answer, and responses are automatically collated against the correct answer; the pertinent prompts and details are furnished in Appendix D.1. For **open-ended evaluation**, models are prompted to generate free-form answers, which are subsequently adjudicated by a large language model judge, with prompts and details available in Appendix D.2. To

Table 3: Open-ended evaluation accuracy (%) (↑) on SPATIALAB-OPEN, disaggregated by question category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline							
	73.53	50.19	70.30	69.81	65.48	62.87	64.93

ascertain the concordance between the LLM judge and human raters, we additionally conducted an LLM-human agreement analysis, as detailed in Appendix D.2. Table 5 attests that the LLM judge (Gemini-2.5-Flash) attains substantial agreement with human annotators, registering a Cohen’s kappa of 0.738 against the majority vote and 0.681–0.795 against individual annotators. Raw accuracy against the majority vote stands at 0.880, whereas Fleiss’ kappa among the human annotators reaches 0.774, together indicative of strong reliability. For supplementary details concerning the experimental setup, prompts, and evaluation procedures, we refer the reader to Appendix D.

Improving Visual Reasoning Capabilities. With a view to enhancing spatial reasoning on our diverse benchmark, we examine an array of reasoning strategies, encompassing the inherent reasoning resident within the VLMs (Appendix H.1), Chain-of-Thought (CoT) prompting (Appendix H.2), CoT augmented with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and multi-agent systems (Appendix H.5). In the case of SFT specifically, Qwen-VL2.5-3B-Instruct was fine-tuned on 40% of the dataset under stratified sampling, thereby improving generalization across relational and geometric tasks. Evaluation on the residual 60% evidenced diminished errors in navigation and orientation, with additive gains accruing from the integration of CoT and SFT. Agentic reasoning, however, revealed that while orientation benefits substantially, the remaining categories either stagnate or deteriorate, an observation that underscores the necessity of deeper perceptual and spatial grounding beyond mere multi-step reasoning.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 sets forth model performance under multiple-choice evaluation on SPATIALAB-MCQ. We observe pronounced variation across models, tasks, and families: overall accuracies span roughly 30–55% (random choice = 25%), whereas the human baseline stands at 87.57%, a disparity that betokens considerable headroom. InternVL3.5-72B, an open-source model, emerges as the strongest performer, at 54.93% accuracy. Reasoning- and instruction-tuned

models (e.g., o4-mini at 53.21%, Gemini-2.5-Flash-Thinking at 52.93%, and GPT-5-mini at 54.29%), together with certain very large open-source models (e.g., InternVL3.5-72B at 54.93% and Llama-3.2-90B at 50.36%), cluster near the summit; scale alone, however, proves non-determinative, as Llama-3.2-11B attains a mere 30.50% notwithstanding its substantial size. Considered category by category, Orientation and 3D Geometry constitute the loci at which many strong models excel, several surpassing 60% on one or the other, whereas Spatial Navigation, Depth & Occlusion, and Size & Scale prove more variable and, on the whole, more refractory. Notably, the spatially specialized models (SpaceOm, SpaceThinker, SpaceQwen) attain only the low-to-mid forties overall, whence specialization does not, of itself, translate into superior MCQ accuracy across our diverse question set. Depth & Occlusion and Size & Scale likewise yield mixed outcomes, with some architectures handling occlusion tolerably well while others founder, a divergence that points to disparate inductive biases or training data. These patterns intimate that instruction-tuning and reasoning capability confer particular benefit upon geometric and orientation reasoning, whereas multi-step grounding tasks such as navigation persist as an open weakness.

SPATIALAB-OPEN. The open-ended evaluation scores presented in Table 3 lie substantially beneath their MCQ counterparts and exhibit a broad dispersion: overall accuracy ranges from approximately 9.6% (InternVL3.5-1B) to 40.9% (GPT-5-mini), while the human baseline resides at 64.9%, a configuration that discloses a considerable gap yet to be bridged. Proprietary models generally head the leaderboard on open-ended tasks (e.g., GPT-5-mini at 40.93%, GPT-4o-mini at 26.00%, and Gemini-2.5-Flash at 30.93%); the reasoning-tuned variants, however, prove especially conspicuous, in that o4-mini (37.86%) and Gemini-2.5-Flash-Thinking (32.77%) markedly outstrip many of their baseline open-source counterparts, a pattern suggestive of the substantial improvement that instruction- and CoT-style tuning confers upon generative outputs. Most small open-source models congregate at the foot of the table (many languishing in the teens; InternVL3.5-1B = 9.64%, Qwen-VL2.5-3B = 12.93%), whereas certain very large open models advance into the low twenties (e.g., InternVL3.5-72B = 23.36%, Qwen-VL2.5-72B = 24.64%), demonstrating that scale alone yields but modest gains. The spatial-specialist models (SpaceOm, SpaceThinker, SpaceQwen) attain merely ~10–13% overall, whence architectural or task specialization does not, of itself, translate into robust open-ended generation across our diverse question set. Across subtasks, the leading models perform comparatively better on orientation and on certain relative-position items (e.g., GPT-5-mini: Rel.Pos = 49.53%, Ori = 37.13%), whereas depth & occlusion, size & scale, and above all spatial navigation remain consistently arduous (many models scoring <30% on these subtasks). This pattern corroborates the earlier finding that multi-step grounding and perceptual reasoning constitute the principal bottlenecks for open generative answers.

5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

Averaged across 25 models, the MCQ→open-ended performance gap in SPATIALAB amounts to 23.0% ($\sigma = 5.5\%$), with subtask means ranging from 22.89% (spatial navigation) to 24.57% (3D geometry). Specialist spatial reasoning models manifest the widest gaps (approximately 27%), most conspicuously in spatial navigation (up to 36.68%) and orientation (34.44%), whereas reasoning-oriented models incur narrower gaps (around 19%) attended by lower subtask variance, a pattern that reflects the stabilizing influence of instruction-tuning and CoT decoding. Correlational analysis confirms that spatial navigation predominates in the overall disparity (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) likewise contributing. Negative or near-zero gaps (e.g., Llama-3.2-11B at -1.98 in depth & occlusion, and o4-mini at -3.18 in relative position) betray the propensity of poorly designed MCQ distractors to misrepresent genuine competence. We hypothesize that this format sensitivity issues from the structural advantage intrinsic to MCQ, from a specialization bias toward categorical tasks, from the sequential-reasoning demands peculiar to spatio-navigation, and from calibration disparities in generative output. Collectively, these findings intimate that MCQ, taken in isolation, is liable to overestimate practical spatial reasoning ability. We accordingly recommend supplementing MCQ with open-ended evaluations, auditing distractors, adopting stepwise generation or instruction-tuning, and reporting per-subtask diagnostics, so as to capture true model performance and attenuate format-dependent artifacts. Further details are provided in Appendix G and Table 18.

5.3 ERROR ANALYSIS

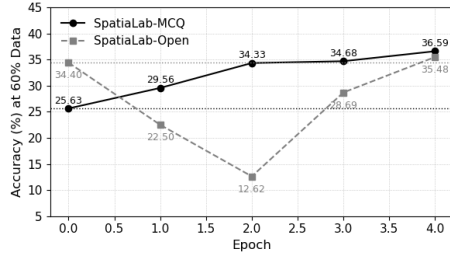
SPATIALAB-MCQ. Our analysis (Appendix E) reveals that VLMs attain selective peaks yet remain bereft of holistic spatial competence. Closed-source models achieve the highest scores (e.g., 85.71% in *Stacking Orientation*), only to collapse to near-chance in *Relative Size Comparison*. Open-source scaling elevates the ceilings (80.56% in *Corner/Angle Positioning*) but does not avert catastrophic failures (2.0% in *Object Rotation*). Reasoning-augmented and spatially specialized models contribute localized gains, as in occlusion inference or navigation, yet remain capped beneath 55% across many tasks of physical abstraction (*Gravity Effects*, *Stability Prediction*). The error distributions further expose systematic weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cues (*Reflective Surfaces*), while the comparatively stronger results in *Obstacle Avoidance* intimate shortcut exploitation rather than robust planning. Considered in aggregate, these patterns confirm that contemporary VLMs lean heavily upon surface-level correlations and want stable encodings of orientation, physics, and compositional logic. Progress will in all likelihood demand both richer training distributions (e.g., embodied and physics-driven data) and architectural mechanisms dedicated to geometric and reference-frame grounding.

SPATIALAB-OPEN. The error analysis of SPATIALAB-OPEN (Appendix F) reveals that even the top-performing closed-source models (e.g., GPT-5-mini at 58.14% in *Directional Relations*) collapse on tasks such as *Proximity Gradients*, an outcome that establishes the task-specific character of their dominance. Large open-source models achieve strong results in *Depth & Occlusion* (e.g., 60.0%) yet nonetheless subside to 9.3% on *Proximity*, demonstrating that scale alone cannot secure robustness. Smaller models exhibit isolated strengths (e.g., 26.0% in *Relative Size Comparison*) but frequently break down in their entirety (e.g., 0.0% in *Betweenness*), a pattern that reflects the limitations inherent in compact architectures. Reasoning-tuned systems evince promise, Gemini-2.5-Flash-Thinking reaching 75.0% in *Tool Handedness*, yet cognate models descend to single digits, thereby laying bare the fragility of their integration. The specialized spatial models fare worst overall, seldom exceeding 20% and repeatedly failing on core relational tasks. The failure patterns foreground systemic weaknesses in occlusion handling, orientation, and multi-step relational chaining, whereas the partial success attained on size-scale cues bespeaks a reliance on superficial correlations. These results implicate three root causes, namely the absence of representations for spatial analogies, architectural limits bound up with scale and training diversity, and brittle reasoning pipelines, and thereby underscore the requirement for unified geometric encodings, physics-aware reasoning, and embodied data if the field is to advance from isolated peaks toward consistent spatial competence.

Qualitative Error Analysis. In order to apprehend model weaknesses more precisely, we conducted a qualitative error analysis (see Appendix I) and found that failures coalesce into a small number of recurring classes rather than constituting random noise. The prevalent errors comprise spatial mislocalization, in which models conflate referents amid crowded scenes; perspective and scale mistakes, in which reliance upon object-size priors supersedes image-based cues; and occlusion or ordering failures, especially where thin or partially concealed structures are concerned. We observe, in addition, attribute confusion, whereby perceptual properties are conflated with functional ones, and open-ended rationalizations that generate fluent yet visually ungrounded narratives. These errors recur consistently across architectures and prompting styles, a regularity that implicates structural biases endemic to contemporary VLMs rather than dataset-specific noise. Significantly, failure rates spike whenever multiple cues must be fused, as when depth ordering must be combined with relative size, and confidence calibration proves especially deficient in open-ended settings. Our diagnostic protocols confirm that models frequently disregard minimal yet decisive visual features, resort to brittle heuristics, and fail to revise their outputs when counterfactual edits are introduced. The root causes implicate insufficient object-centric binding, a paucity of geometric supervision, and training objectives that reward plausibility at the expense of grounding. Taken together, these findings accentuate that contemporary VLMs achieve strong coarse perception yet struggle with multi-cue integration and grounded reasoning, thereby underscoring the need for geometry-aware supervision, multi-scale feature retention, and verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. We observe that reasoning-enabled models consistently surpass their baselines in both MCQ and open-ended formats, the most substantial gains accruing to relational and orientation tasks (e.g., +13.1% in *Relative Positioning* for MCQ). Although reasoning serves



Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(a) Trends in SFT performance over the training epochs. (b) A comparison of performance (before versus after) across categories for the MCQ and Open-ended formats.

Figure 4: SFT training results: (a) the learning trends over the epochs, and (b) the final accuracy values.

to stabilize open-ended performance, the improvements remain uneven across categories, declining most notably in Size & Scale, an unevenness that underscores that reasoning modules bolster logical consistency without wholly resolving grounding or scale sensitivity. Detailed per-category comparisons are furnished in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Across models, CoT prompting confers scant benefit and frequently depresses accuracy (Table 19), orientation being the sole category to register consistent gains. This pattern indicates that, while step-by-step reasoning aids directional alignment, it founders on tasks that demand robust perceptual grounding, such as depth, scale, and spatial navigation. In contradistinction to textual reasoning, CoT here tends to amplify flawed priors rather than to rectify them, an observation that reinforces the contention that solving SPATIALAB presupposes perceptual understanding beyond mere logical chaining (details in Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. The augmentation of CoT prompting with a self-reflection stage, wherein models are explicitly prompted to review their antecedent reasoning step by step and to redress errors (see Figure 8 in the Appendix for the prompt), yields modest benefits in multiple-choice settings, most appreciably for geometry and depth. This benefit, however, fails to generalize to open-ended tasks (Appendix H.3), a failure that confirms the incapacity of reflective reasoning to compensate for absent perceptual grounding and thereby brings to the fore a core limitation of contemporary VLMs.

Supervised Fine-Tuning (SFT). As displayed in Figure 4, supervised fine-tuning (SFT), conducted on Qwen2.5-VL-3B-Instruct by means of a stratified 40% split of SPATIALAB (~560 samples), consistently improves MCQ accuracy across every spatial reasoning category. It nonetheless affords little transfer, and at times negative transfer, to open-ended tasks. This divergence is symptomatic of overfitting to task-specific answer distributions, and thereby reinforces the concern that SFT frequently *teaches to the test* rather than cultivating generalizable reasoning (Wang et al., 2025). The precipitous decline in open-ended performance is indicative of biased internal representations, finely tuned for categorical discrimination yet unstable in generative settings. A probable cause resides in the catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a), compounded by objectives that privilege discrete-choice alignment over spatial reasoning expressed in language. On balance, then, although SFT enhances performance within structured formats, it courts the imposition of an algorithmic straitjacket, elevating accuracy on constrained tasks while stalling progress on naturalistic spatial reasoning. Further details are available in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). In order to probe agentic capabilities, we deploy SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) and devised to perform structured spatial reasoning over images. This framework decomposes complex visual reasoning into specialized sub-tasks apportioned to dedicated agents, among them modules for object segmentation, attribute extraction, spatial-relation mapping, and transformation tracking. By consolidating the resulting outputs into a unified structured representation, the system endeavors to propagate low-level perceptual cues through higher-order relational inferences. We implemented the pipeline using Gemini-2.5-Flash at low temperature so as to secure deterministic outputs throughout the multi-stage refinement process. The experimental results evidence substantial gains within the Orientation category, an improvement of +8.00% in MCQ and a striking +36.00% in open-ended evaluation. This success, however, is by no means universal: categories such as Depth & Occlusion and Spatial Navigation sustained severe declines of -24.00% and -12.00%

in open evaluation, respectively. Such regressions indicate that, wherever the inherent perceptual priors, depth or occlusion cues among them, are weak, multi-step reasoning is prone to entrench misconceptions rather than to correct them. Ultimately, these findings intimate that, although agentic workflows enhance tasks reducible to sequential alignment, they cannot offset a deficit of fundamental perceptual grounding in complex 3D environments. Further details are available in Appendix H.5.

6 DISCUSSION

Spatial reasoning is integral to embodied intelligence, and yet the preponderance of benchmarks continues to rely upon synthetic scenes, templated questions, or narrowly circumscribed task scopes that overlook real-world visual variability. We introduce SPATIALAB, a benchmark of 1,400 visual question-answer pairs encompassing six major spatial categories and 30 subcategories, 3D geometry, occlusion, and navigation among them. The benchmark employs both Multiple-Choice and Open-Ended formats in order to dissociate option-based pattern exploitation from genuine generative reasoning. Its quality is buttressed by a multi-stage review process together with complexity checks that span lighting, texture, and spatial diversity. Evaluations of more than 25 contemporary models alongside human baselines demonstrate that SPATIALAB operates as a precise diagnostic instrument for isolating the weaknesses that beset current vision-language systems.

Our detailed error analysis uncovers systematic failures in the treatment of transparency, multi-step navigation, and occlusion, an indication that models frequently subsist upon surface-level texture cues rather than upon internalized three-dimensional geometric representations. Whereas interventions such as Supervised Fine-Tuning (SFT) yielded unambiguous gains on structured MCQ tasks, they concomitantly precipitated instability in open-ended reasoning, a symptom of the “catastrophic forgetting” of linguistic priors that attends optimization for discrete answers. Reasoning-based approaches such as Chain-of-Thought, moreover, proved efficacious for sequential tasks like Orientation but failed to redress fundamental perceptual errors in Depth and Scale, more often amplifying hallucinations than resolving them. Agentic decompositions exhibited analogous trade-offs, ameliorating specific alignment tasks while degrading performance in contexts demanding holistic scene understanding. These divergent behaviors jointly confirm that scaling laws and reasoning scaffolds are, in themselves, insufficient; the attainment of human-level spatial intelligence will demand architectural innovations that explicitly integrate geometric grounding and physics-aware representations.

7 CONCLUDING REMARKS

Spatial reasoning persists as a cardinal bottleneck for VLMs, most acutely within visually complex, real-world settings. SPATIALAB introduces 1,400 diverse QA pairs distributed across six spatial categories and thirty subcategories, thereby enabling fine-grained evaluation by way of both MCQ and open-ended formats. Notwithstanding the application of several performance-enhancing methods, our experiments disclose substantial gaps between state-of-the-art models and human performance, open-ended generation incurring an additional 10–25% degradation relative to multiple-choice accuracy. The error analysis reveals persistent failures in depth, occlusion, 3D geometry, and multi-step navigation, a pattern that reflects a reliance upon superficial cues in preference to grounded spatial representations. SPATIALAB lays these limitations bare and furnishes a structured testbed for the diagnosis of spatial competence. By enabling holistic evaluation, our benchmark furnishes actionable insights for the development of VLMs endowed with human-aligned spatial understanding. Future progress will demand richer training signals (e.g., embodied interaction and action-conditioned feedback) alongside architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We have undertaken multiple measures to ensure ethical compliance throughout the development of this work. All metadata associated with the images were expunged so as to safeguard privacy, and any personally identifiable features (such as human faces or license plates) were minimized or excluded wherever practicable. The dataset was assiduously curated to obviate copyright concerns, with diligent

attention accorded to sourcing and licensing, notwithstanding the possibility of minor oversights despite our best efforts. The benchmark construction process was conceived with transparency and fairness foremost in view, thereby ensuring that no demographic group was singled out or adversely represented. We adhered to the ICLR Code of Ethics and affirm that neither human subjects nor sensitive personal data were implicated.

REPRODUCIBILITY STATEMENT

In order to facilitate reproducibility, we provide a detailed exposition of the dataset construction process, the annotation pipeline, and the evaluation protocol across both the main text and the appendix. All filtering, quality-control, and validation steps are explicitly documented, accompanied by unambiguous definitions of the task categories and annotation criteria. The hyperparameters, model configurations, and training procedures are reported in the appendix in order to underwrite the consistent replication of our results.

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A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning, the capacity to apprehend geometric structures and relational layouts within visual scenes, is integral to embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) interrogated compositional reasoning by means of synthetic or curated templates, an approach that permitted controlled analysis while conferring only limited ecological validity. With the advent of large vision–language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023), spatial reasoning has been revisited within more naturalistic datasets, though systematic weaknesses persist. Notwithstanding their strong performance on captioning and visual QA, these models frequently falter on tasks that require depth ordering, occlusion inference, or consistent reference-frame alignment. This discrepancy throws into relief a more encompassing problem, namely that scaling alone does not warrant progress in structured spatial understanding.

Spatial reasoning benchmarks. A spectrum of benchmarks has of late been proposed with a view to probing these gaps (Table 1). SCANQA (Azuma et al., 2022) figured among the earliest embodied QA datasets, exacting reasoning about 3D indoor environments, yet its circumscribed scale (800 scenes) and comparatively shallow complexity curtail broader generalization. Datasets such as VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) broadened their purview to real-world or household contexts, but their reliance upon template-driven questions engendered attenuated linguistic variability together with an overrepresentation of simple binary or multiple-choice relationships. More recent undertakings, among them EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), augmented task variety while remaining constrained by scale (2.2K and 211 samples, respectively) and by annotation style (template-heavy, or else manual but narrow). SPATIALRGPT-BENCH and BLINK-SPATIAL introduced mixed domains and as many as 14 categories, and thereby enhanced diversity, yet continued to emphasize either static snapshots or limited multi-view simulation.

Embodiment, difficulty, and diversity. The movement toward embodiment is discernible in datasets such as ROBOSPATIAL (Song et al., 2024), which scales to as many as 1M scenes furnished with robot-centric questions. Its template-based annotation, however, renders the questions less natural and the more readily gameable through heuristic shortcuts. In like manner, SPATIALVLM (Chen et al., 2024b) offers unprecedented scale (billions of QA pairs), yet its coverage of merely two categories renders it unsuitable for fine-grained diagnostic evaluation. At the opposing extreme, high-quality manual datasets such as VSI-BENCH (Yang et al., 2024) attain improved annotation fidelity but, encompassing only 288 scenes, remain inadequate for training or for robust cross-model comparison. As regards complexity, the majority of extant benchmarks are capped at “medium” (Mm in Table 1), and seldom extend to the “hard” or “extremely hard” strata that necessitate multi-step causal inference. A wide evaluation gap is thereby left open for next-generation models.

Evaluation protocols. Of comparable importance are the evaluation type and the depth of analysis. The bulk of prior work adopts close-ended QA or binary judgments (e.g., VISUAL SPATIAL, EMBSPATIAL-BENCH), which streamline scoring at the cost of constraining the expressivity of model outputs. A handful of datasets, among them ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN, introduce open-ended evaluation, yet only SPATIALAB-OPEN couples it with exceedingly high complexity, manual puzzle-style annotations, and a comprehensive analysis of errors. This conjunction affords a more penetrating understanding of systematic weaknesses beyond accuracy, calibration and cross-model divergence included, which prevailing datasets seldom capture. In the absence of such granularity, apparent performance gains may prove misleading, inasmuch as models are liable to overfit to dataset-specific biases rather than to generalize robust spatial competence.

Positioning of SPATIALAB. SPATIALAB confronts these limitations directly. By comparison with prior datasets, which span between 2 and 14 task categories, SPATIALAB traverses 30 distinct categories across geometry, occlusion, orientation, relative positioning, size and scale, and navigation, and thereby affords unprecedented task diversity. In contradistinction to template-based benchmarks, all of its questions are manually authored and puzzle-inspired, a provenance that secures linguistic richness while curtailing the scope for shortcut exploitation. SPATIALAB is, furthermore, fully embodied and multi-modal, subsuming both static and dynamic visual contexts, and thus bridges the

divide between robotics-style datasets (ROBOSPATIAL) and web-scale corpora (SPATIALVLM). As Table 1 attests, even the strongest baselines, InternVL3.5-72B and GPT-5-mini among them, perform well beneath 60%, the open-ended variant yielding a mere 40.93% accuracy. This pronounced performance gap at once evidences the difficulty of the benchmark and its utility for propelling future research.

Toward next-generation evaluation. By conjoining high difficulty, broad coverage, manual annotation, and open-ended tasks, SPATIALAB extends the landscape of spatial reasoning benchmarks from template-driven, narrow evaluations toward comprehensive, embodied testing. This design renders it feasible not merely to adjudicate whether a model answers correctly, but further to identify systematic error patterns, to probe causal and counterfactual reasoning, and to gauge calibration reliability. In this respect, SPATIALAB constitutes a rigorous foundation for the ensuing phase of multimodal research, complementing datasets such as MIND THE GAP and VLM4D that explore temporal or generative facets while remaining narrower in coverage. We anticipate that future work will harness SPATIALAB as a high-bar diagnostic instrument, impelling models beyond surface-level recognition toward genuinely grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

In order to situate SPATIALAB within the broader landscape, we conducted a detailed examination of existing spatial reasoning benchmarks. Earlier efforts have contributed important foundations; they nonetheless exhibit structural and methodological constraints that curtail their value for the evaluation of current VLMs. These constraints fall into three principal areas, namely synthetic-domain bias, limited task scope, and restrictive evaluation formats. Table 4 recapitulates the major limitations we identify.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA are predicated upon synthetic environments or template-generated linguistic structures. Although these settings afford controlled conditions conducive to isolating compositional reasoning, their visual domain wants real-world complexity and their linguistic space is narrow. Models may consequently subsist upon template regularities in lieu of grounded spatial interpretation. Cognate simulated benchmarks such as ROBOSPATIAL labor under analogous difficulties, which issue in a pronounced sim-to-real gap.

Domain and Scale Limitations. Datasets grounded in 3D reconstruction, ScanQA and Space3D-Bench among them, supply depth supervision yet remain confined to indoor scenes, a confinement that precludes assessment within the outdoor or mixed environments indispensable to embodied agents. High-quality manual datasets such as VSI-Bench offer richer annotation but remain modest in scale, a limitation that constrains their usefulness for high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM span broad tasks yet lack detailed taxonomies, a deficiency that renders them unsuitable for diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, EMBSPATIAL-BENCH and WHAT’S UP included, lean heavily upon Binary or MCQ formats. Such formats are apt to inflate measured performance by countenancing option elimination in place of genuine generative spatial reasoning. SPATIALAB mitigates these shortcomings by conjoining MCQ and Open-Ended formats across a detailed taxonomy of 30 spatial subcategories.

Table 4: Limitations identified in representative prior spatial benchmarks, set against SPATIALAB. The limitations are categorized according to domain constraints, evaluation format, and task taxonomy.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain; template-driven language; prone to shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Restricted to indoor point clouds; limited scale (< 1K scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a small number of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats; limited complexity in reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes produce a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~200 samples) prevents robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Emphasizes metric outputs; limited support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Addresses identified limitations with real-world images, detailed taxonomy, and diverse evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

The comprehension of spatial reasoning demands considerably more than the mere recognition of objects; it hinges upon the manner in which those objects relate to, align with, and interact with one another in physical space. In order to capture this complexity, we introduce *SpatiaLab*, a benchmark constructed to interrogate fine-grained facets of spatial understanding across a diverse array of categories. The tasks constitutive of SpatiaLab mirror the ways in which humans naturally describe and reason about environments, ranging from relative positioning (e.g., *What’s to the left of the microwave?*) to geometric inference (e.g., *Would this stack fall if another book is added?*). Each category is further decomposed into subcategories that foreground subtle yet essential dimensions of reasoning, occlusion, orientation, navigation, and physical effects such as gravity among them. By framing every subtask in accessible, everyday terms, SpatiaLab impels models to transcend object detection and to engage in structured spatial interpretation. This design enables us to ascertain how effectively systems generalize across the contexts that bear upon robotics, AR/VR, autonomous driving, and human–computer interaction. The sections that follow furnish concise, task-level descriptions for every category and subcategory in SpatiaLab.

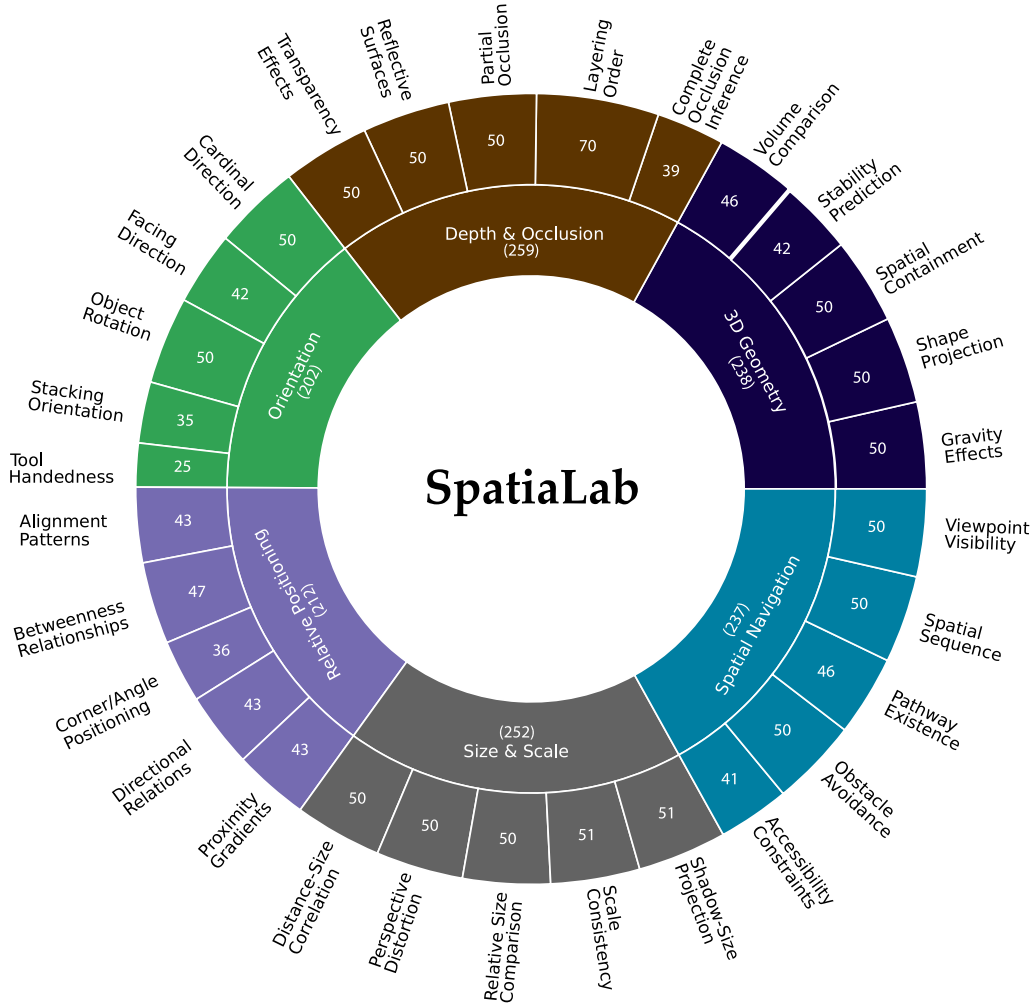


Figure 5: **Categories and subcategories of spatial reasoning in SPATIALAB.** Each category is decomposed into five subcategories, thereby yielding thirty task types in the aggregate.

The following delineates the rationale and real-world utility that inform each task design, presented category by category:

1296 B.1 RELATIVE POSITIONING

1297
1298 **Directional Relations.** Directional relations characterize the manner in which objects are positioned
1299 relative to one another, whether to the left, right, above, or below. A command of these relations en-
1300 ables models to interpret spatial layouts more naturally, much as humans describe scenes; identifying
1301 what lies immediately to the left of an appliance, for instance, subserves scene understanding and
1302 robot navigation.

1303 **Proximity Gradients.** Proximity reasoning entails discerning which objects are nearest to, or most
1304 remote from, a reference point. This capacity is pivotal for questions such as “Which object is nearest
1305 to the doorway?” Models capable of estimating gradients of closeness can underwrite navigation
1306 tasks and assistive technologies.

1307 **Alignment Patterns.** Alignment concerns whether objects conform to a particular orientation or
1308 ordering, such as being stacked vertically or arrayed in a straight line. The recognition of alignment
1309 patterns facilitates an understanding of organization within environments such as shelves, tables, or
1310 warehouses, and is, moreover, critical for tasks such as robotic sorting and automated inspection.

1311 **Betweenness Relationships.** Betweenness centers upon identifying the objects that occupy the
1312 middle ground between two references. To ask what sits between the lamp and the armchair, for
1313 instance, is to demand reasoning about spatial continuity, which in turn subserves navigation, retrieval,
1314 and situational awareness.

1315 **Corner/Angle Positioning.** Corners and angled positions frequently function as reference anchors in
1316 spatial reasoning. Questions such as “What’s positioned in the back-right corner?” require the model
1317 to localize objects within constrained regions, a mode of reasoning that proves vital in robotics for
1318 the purposes of mapping and placement.

1319 1320 1321 B.2 DEPTH & OCCLUSION

1322
1323 **Layering Order.** Layering requires the disambiguation of which objects appear in front and which
1324 behind. Identifying the frontmost item on a table, for example, is an exercise in depth reasoning, one
1325 of particular utility in AR/VR applications wherein visual stacking conditions the user experience.

1326 **Partial Occlusion.** Partial occlusion arises when one object conceals part of another from view. To
1327 recognize the item hidden behind a coffee cup, for instance, is to reconstruct missing information, a
1328 capacity that fortifies the robustness of perception systems operating in cluttered environments.

1329 **Complete Occlusion Inference.** On occasion objects are wholly concealed, and yet cues such as
1330 shadows or bulges betray their presence. To infer what lies beneath a blanket obliges the model to
1331 marshal indirect evidence, a form of reasoning that mirrors human inference under uncertain visual
1332 conditions.

1333 **Transparency Effects.** Transparent surfaces, such as glass or water, permit the partial visibility
1334 of objects situated behind them. To apprehend what is visible through a fish tank requires the
1335 amalgamation of occlusion reasoning with transparency perception, a competence that assumes an
1336 important role in simulation, graphics, and embodied AI.

1337 **Reflective Surfaces.** Reflective reasoning charges models with interpreting mirrors, lustrous metals,
1338 and glass reflections. To identify what is reflected in a mirror entails indirect spatial mapping, a
1339 faculty indispensable to safety in autonomous vehicles and to situational awareness in robotics.

1340 1341 1342 B.3 ORIENTATION

1343
1344 **Cardinal Direction.** Cardinal reasoning pertains to the compass directions, north, south, east, and
1345 west. To ask which way a front door faces is to require the integration of global orientation cues, a
1346 capacity critical to navigation and geospatial alignment.

1347 **Object Rotation.** Rotation reasoning quantifies the extent to which an object has been turned about
1348 an axis. Estimating that a chair has been rotated by some number of degrees proves crucial for
1349 accurate 3D modeling, and assists robots in aligning objects correctly for manipulation.

1350 **Facing Direction.** Facing direction specifies the orientation of an object's front face. Determining
1351 whether a monitor faces north, for example, requires mapping its surface orientation onto global
1352 references, a matter of consequence in surveillance and robotics.

1353 **Stacking Orientation.** Stacking concerns the discernment of whether objects are piled vertically,
1354 horizontally, or in mixed orientations. Recognizing such layouts guarantees stability in warehouses
1355 and homes and is, in addition, critical to safe robotic stacking.

1356 **Tool Handedness.** Certain tools, scissors and gloves among them, are designed for left- or right-
1357 handed use. To identify handedness requires fine-grained perception of shape and orientation, a
1358 competence that augments assistive robotics within human-centered environments.

1359

1360

1361 B.4 SIZE & SCALE

1362

1363 **Relative Size Comparison.** Relative comparisons adjudicate whether one object is larger, taller, or
1364 smaller than another. To ask whether a vase is taller than a statue is to require proportional reasoning,
1365 a faculty crucial to packing, fitting, and recognition tasks.

1366 **Perspective Distortion.** Perspective can render distant objects apparently smaller than proximate
1367 ones, even where the actual sizes diverge. To identify illusions such as a faraway car appearing
1368 smaller than a nearby bicycle requires compensation for depth cues, an ability that renders models
1369 more robust to camera perspective.

1370 **Distance-Size Correlation.** Here size judgments are contingent upon the estimation of distance.
1371 A nearby bicycle appearing larger than a distant car, for instance, ought not to confound reasoning
1372 about true size, a discrimination that assists navigation, AR/VR, and visual grounding.

1373 **Scale Consistency.** Scale reasoning entails verifying whether object sizes are plausible in context. A
1374 mouse larger than a chair signals an inconsistency that models must detect, a capacity that reinforces
1375 realism in simulated environments.

1376 **Shadow-Size Projection.** Shadows afford indirect cues to object size and to the position of the
1377 light source. To estimate how tall a light must be on the basis of shadow length requires geometric
1378 reasoning, of use in graphics, architecture, and physics-based modeling.

1379

1380

1381 B.5 SPATIAL NAVIGATION

1382

1383 **Pathway Existence.** Pathway reasoning ascertains whether a clear route obtains between two points.
1384 Determining whether one can reach the desk without displacing chairs, for example, tests navigability,
1385 an ability that underpins indoor robot planning.

1386 **Obstacle Avoidance.** Avoidance requires the identification of routes that minimize collisions. To ask
1387 for the clearest path from door to window maps directly onto robotic and human navigation, and is
1388 indispensable to autonomous driving and to indoor mobility aids.

1389 **Viewpoint Visibility.** From a given vantage, certain objects may be occluded while others remain
1390 visible. To ascertain what is obscured when one is seated on the couch, for example, requires the
1391 simulation of perspective, a competence that heightens AR/VR realism and robotic vision.

1392 **Spatial Sequence.** Sequence reasoning establishes the order in which entities are encountered along
1393 a path. In walking from the kitchen to the balcony, one might pass the dining table first, a mode of
1394 reasoning that emulates human navigational memory.

1395 **Accessibility Constraints.** Constraints identify which objects are barred from access on account of
1396 obstacles. A cabinet may be unreachable, for instance, by virtue of a table positioned before it, a
1397 consideration crucial to assistive AI and to ergonomic design.

1398

1399

1400 B.6 3D GEOMETRY

1401

1402 **Volume Comparison.** The comparison of volume adjudicates which object occupies the greater
1403 three-dimensional extent. Judging whether a box is larger than a sphere subserves packing and
containment tasks, a capacity of importance to logistics and simulation.

Stability Prediction. Stability reasoning forecasts whether the addition or displacement of objects will topple a stack. Ascertaining whether placing a book will collapse a pile, for instance, mirrors human intuition and thereby underpins safety in robotics and construction.

Shape Projection. Projection tasks pose the question of what shape an object would cast under specified rotations or illumination. A sphere invariably projects a circle, for example, whereas a cube may project a square or a hexagon, and reasoning of this kind fortifies geometric understanding.

Spatial Containment. Containment verifies whether one object can be accommodated within another. Reasoning about whether a ball can fit inside a box, for example, subserves packing and design, and aligns with intuitive geometry.

Gravity Effects. Gravity reasoning predicts the outcomes of motion under physical forces. A rolling ball may come to rest beneath a table on account of friction and intervening barriers, for instance, and reasoning of this sort augments real-world simulation capabilities.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

We recruited annotators aged between 24 and 30 years, all of them engineering graduates, who participated voluntarily and labored independently without external incentive. Prior to the commencement of annotation, we furnished structured training sessions in order to secure clarity as to the task requirements and consistency across annotators, and we upheld all ethical and quality-control standards throughout the process. This approach ensured that the resulting annotations were at once reliable and congruent with the dataset design.

C.2 ANNOTATOR TRAINING AND PROTOCOL

Prior to the inception of annotation, we devised a structured training program in order to secure consistency and reliability across the full taxonomy of tasks. We first acquainted annotators with the six core categories (*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*) and their thirty subcategories. For each subcategory, we supplied curated examples illustrative of the corresponding spatial relations (e.g., directional relations, occlusion inference, stability prediction), together with counter-examples calculated to foreground common pitfalls. Beyond the taxonomy proper, we trained annotators to attend to six meta-dimensions of image variability, namely lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints, so that the questions would explicitly capitalize upon visual diversity. We demonstrated how these factors interact within real-world settings, as in reasoning about occlusion under low-contrast lighting or preserving scale consistency under perspective distortion.

We further introduced the two evaluation modes, multiple-choice and open-ended. Within multiple-choice settings, annotators learned to devise precise distractors so as to probe fine-grained discrimination, whereas within open-ended settings they rehearsed the formulation of prompts conducive to compositional reasoning. In order to ground this training, we conducted calibration sessions in which annotators generated both types of question for sample images and iteratively refined their designs in light of feedback. So as to situate the task design within the broader research landscape, we reviewed prior benchmarks (e.g., SpatialBot-Bench, EmbSpatial, SpatialMM) alongside the annotators and analyzed their limitations. Such resources frequently privilege simplified relations, synthetic or staged scenes, and narrow reasoning types, while neglecting natural complexity of the kind exemplified by shadows, reflections, or gravity-sensitive stability. This gap analysis threw into relief the need for richer, more ambiguous, and dynamically layered spatial contexts.

Finally, we provided detailed demonstrations of the annotation protocol. These demonstrations specified how images of sufficient spatial complexity were to be selected, how questions were to be mapped unambiguously onto subcategories, and how trivial formulations were to be eschewed. We enjoined annotators to prioritize tasks demanding multi-step reasoning, depth layering, relational chains, or cross-view scale inference among them, over binary labels. By way of this systematic training and iterative feedback, we ensured that the final dataset balanced taxonomic coverage against real-world relevance.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

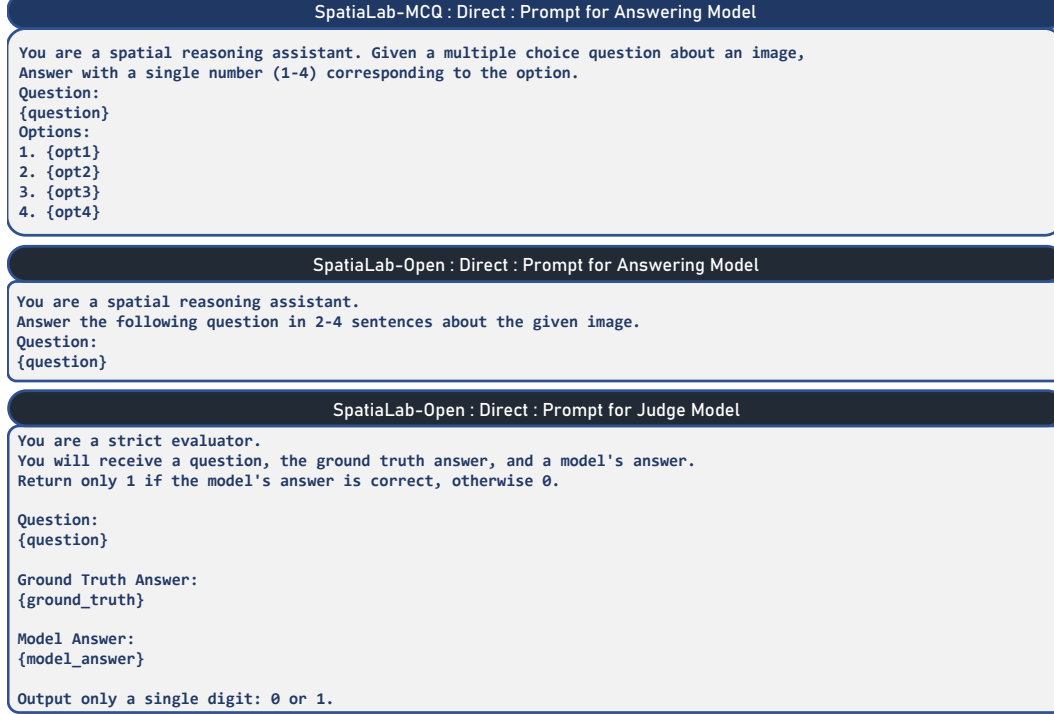


Figure 6: Prompts employed for evaluation.

D.1 MULTIPLE CHOICE EVALUATION

The prompts employed for the answer model are exhibited in Figure 6. We set `temperature` to 0 and `max_tokens` to 10, while retaining all remaining generation parameters at their default values.

D.2 OPEN-ENDED EVALUATION

The prompts employed for both the answer model and the judge model are provided in Figure 6. We set `temperature` to 0.7 and `max_tokens` to 1200, with all other parameters likewise retained at their default values.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

In order to appraise *Gemini-2.5-Flash* in its capacity as an automatic judge, we selected a balanced set of 240 samples spanning all categories. Each sample encompassed the evaluation setup (the question, the ground truth, and the model output), which was scored independently by three human annotators, who assigned binary scores, 1 for correct and 0 for incorrect. We collated the LLM’s judgments against the human annotations by means of multiple complementary agreement metrics, computed both in aggregate and per category. Specifically, we employed **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**, defined as follows.

Cohen’s Kappa. Cohen’s kappa (κ) quantifies the pairwise agreement between two raters while correcting for the agreement attributable to chance. For a binary task, it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, wherein p_o denotes the observed agreement and p_e the agreement expected by chance: $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here $p_c^{(1)}$ and $p_c^{(2)}$ designate the marginal probabilities with which each rater assigns class c . An $\kappa = 1$: perfect agreement; $\kappa = 0$: agreement equals chance; $\kappa < 0$: systematic disagreement) We computed Cohen’s kappa between the LLM and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}), as well as between the LLM and the annotators’ majority vote (κ_{majority}).

Accuracy with Majority Vote. We further measured the raw agreement between the LLM and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, wherein y_i^{LLM} denotes the LLM’s decision for sample i and y_i^{maj} the majority vote among the annotators. In contradistinction to Cohen’s kappa, accuracy effects no correction for chance agreement; it merely reflects the proportion of identical labels.

Fleiss’ Kappa. In order to assess inter-annotator reliability, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa to the case of multiple raters. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, wherein n is the number of annotators. The observed agreement is $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, and the expected agreement is $\bar{P}_e = \sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa is thereupon computed as $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Higher values betoken stronger inter-annotator reliability beyond chance.

Table 5: Agreement metrics between Gemini-2.5-Flash, in its capacity as an automatic judge, and three human annotators across all categories. The table reports Cohen’s kappa against both the majority vote and each individual annotator, raw accuracy against the majority, and Fleiss’ kappa among all annotators. Bold values denote the highest agreement within each category.

Category	κ Majority	κ Human1	κ Human2	κ Human3	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Interpretation Table 5 evidences that the LLM judge (Gemini-2.5-Flash) exhibits strong overall agreement with the human annotators, attaining a Cohen’s kappa of 0.738 against the majority vote and 0.681–0.795 against individual annotators. Raw accuracy against the majority stands at 0.880 overall, while Fleiss’ kappa among the human annotators reaches 0.774, an indication of substantial consistency. Certain categories, 3D Geometry among them, attain perfect agreement, with $\kappa = 1.000$ and an accuracy of 1.000. Depth & Occlusion likewise exhibits high agreement ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909). Categories such as Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714) evince moderate agreement, yet remain within acceptable reliability ranges.

These results indicate that Gemini-2.5-Flash performs on a par with human evaluators in the scoring of correctness, its Cohen’s kappa values residing for the most part in the substantial range (0.61–0.80) and certain categories approaching almost perfect agreement (in excess of 0.81). The Fleiss’ kappa of 0.774 confirms that the human annotators are themselves highly consistent, thereby validating the reliability of the reference scores. The LLM attains perfect or near-perfect agreement in clearly delimited categories such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933), a testament to accurate pattern recognition. The moderate agreement observed in categories such as Orientation and Size & Scale reflects task complexity and inherent subjectivity rather than any shortcoming of the model. On the whole, then, the LLM-as-judge demonstrates robust reliability and may serve as a practical automated scoring tool that closely mirrors human evaluation.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. Closed-source models (Table 6) yield some of the strongest results across the benchmark. The foremost instance is **GPT-4o-mini**, which attains 85.71% in *Stacking Orientation*, the highest score overall. **Gemini-2.0-Flash** performs creditably in occlusion and projection, achiev-

Table 6: Multiple-choice evaluation accuracy on SPATIALAB-MCQ, disaggregated by question sub-category, for the closed-source models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	56.0	42.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

ing 64.0% in *Partial Occlusion* and 64.71% in *Shadow-Size Projection*. In like manner, **GPT-5-mini** scores 68.0% in *Tool Handedness* and 55.71% in *Layering Order*. Weaknesses nonetheless persist, inasmuch as **Claude 3.5 Haiku** attains a mere 32.0% in *Relative Size Comparison*. On average, closed-source systems outstrip their open-source counterparts by 15–25 percentage points across several categories. The within-model variation, however, remains considerable, frequently exceeding 50 percentage points. We therefore conclude that, although closed-source training pipelines confer higher ceilings, they do not resolve the challenge of uniform spatial competence.

Open-source Large Models. Upon scaling to large open-source models (Table 7), we discern manifest improvements alongside severe fragilities. **InternVL3.5-72B** achieves 80.56% in *Corner/Angle Positioning*, among the highest of the open-source results, and both **InternVL3.5-72B** and **llama-3.2-90b-vision-instruct** reach 66.67% in *Complete Occlusion Inference*. Weaknesses nevertheless persist, as when **llama-3.2-11b-vision-instruct** scores a mere 2.0% in *Object Rotation*. Gains are likewise evident in *Stacking Orientation*, wherein **llama-3.2-90b-vision-instruct** scores 71.43%, though this is purchased at the cost of losses in other categories. The within-model spread frequently exceeds 70 percentage points, a signature of brittle specialization, while gravity reasoning remains modest at 56.0%. We find that scaling engenders higher peaks yet fails to resolve structural weaknesses; our conclusion, accordingly, is that size amplifies selective strengths while leaving foundational representational gaps intact.

Open-source Small Models. We observe that small open-source models (Table 8) manifest sharp contrasts in performance across sub-categories. **qwen-2.5-vl-7b-instruct**, for example, reaches 74.29% in *Stacking Orientation*, whereas **InternVL3.5-1B** collapses to 12.0% in *Cardinal Direction*. Scores in *Pathway Existence* remain modest, the best models tied at 34.78%, and gravity-related reasoning proves similarly feeble, with only 50.0% attained by **InternVL3.5-4B** and **Gemma-3-4B-it**. The within-model spread frequently exceeds 60 percentage points, an indication of brittle generalization. These patterns intimate that smaller models are markedly cue-dependent, performing well wherever visual templates are salient but faltering on tasks that demand deeper spatial abstraction. We construe this inconsistency as evidence that representation learning proceeds unevenly across sub-categories. On the whole, we find that small models afford localized competence yet want robustness across the full spectrum of spatial reasoning.

Table 7: Multiple-choice evaluation accuracy on SPATIALAB-MCQ, disaggregated by question sub-category, for the large open-source models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3.5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

Table 8: Multiple-choice evaluation accuracy on SPATIALAB-MCQ, disaggregated by question sub-category, for the small open-source models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

Reasoning Models. Reasoning-enhanced models (Table 9) evince notable benefits on select tasks yet remain inconsistent overall. **o4-mini** reaches 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*, thereby outperforming most open baselines, while **Gemini-2.0-Flash-Thinking** achieves 62.75% in *Shadow-Size Projection* and 52.17% in *Volume Comparison*, a testament to stronger geometric reasoning. The strongest results issue from **Gemini-2.5-Flash-Thinking**, which scores 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*. The same family, however, falters elsewhere, registering 34.0% in *Cardinal Direction* and 34.0% in *Shape Projection*. This variance demonstrates that reasoning mechanisms deepen competence within certain sub-categories without broadening coverage, and the within-model spreads frequently exceed 35 percentage points. We conclude that reasoning augmentation confers localized capacity while leaving structural gaps in spatial understanding intact; on the whole, these models betoken promise rather than generalizable competence.

Table 9: Multiple-choice evaluation accuracy on SPATIALAB-MCQ, disaggregated by question sub-category, for the reasoning models. We bold and underline the best score within each model category.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Spatial Reasoning Models. Spatially specialized models (Table 10) disclose targeted improvements yet want generality. **SpaceQwen2.5-VL-3B-Instruct** achieves 48.0% in *Distance-Size Correlation* and 49.02% in *Scale Consistency*, thereby manifestly surpassing many open-source baselines in size-scale reasoning, and the same model leads on navigation tasks, at 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** reach 50.0% in *Stability Prediction*, in keeping with expectations for domain-specific reasoning. Performance nonetheless declines sharply on other tasks, to a mere 24.0% for *Spatial Containment*, and these models seldom exceed 55.0% even in their strongest domains, the spread across sub-categories frequently surpassing 30 percentage points. We find that specialization drives narrow peaks without producing holistic competence; our interpretation is that domain targeting augments select abilities yet fails to integrate them within a robust, general framework.

Overall Discussion. Across all categories (Table 8, Table 7, Table 6, Table 10, Table 9), we observe consistent fragmentation. Small open-source models peak at 74.29% in *Stacking Orientation* yet collapse to 12.0% in *Cardinal Direction*. Larger models elevate the ceilings, to 80.56% in *Corner/Angle Positioning*, but nonetheless fail catastrophically, at 2.0% in *Object Rotation*. Closed-source systems attain the strongest scores, **GPT-4o-mini** reaching 85.71% in *Stacking Orientation*, and yet subside to 32.0% in *Relative Size Comparison*. Specialized models exhibit strengths in navigation (53.66%) and stability (50.0%) while remaining capped beneath 55% overall, and reasoning models register gains such as 71.79% in *Complete Occlusion Inference* even as others linger at 34.0%. The within-model spreads frequently exceed 50 percentage points, a hallmark of brittle internal consistency, and the sub-categories that demand physical abstraction, *Gravity Effects*, *Stability Prediction*, and *Spatial Containment*, consistently score beneath 35–40% irrespective of category. We ascribe these failures to two root causes, namely the absence of stable encodings of orientation and physics, and an overreliance upon surface-level correlations. The wide variance, spanning highs of 85.71% and lows of 2.0%, intimates that models are pattern-matching rather than reasoning robustly about space. Scaling, proprietary training, and reasoning augmentation each confer benefits, and yet none secures general coverage; each approach instead yields isolated peaks that coexist with glaring blind spots. The collective picture, in short, is one of fragmented competence rather than integrated ability. We conclude that progress in spatial reasoning will demand architectural changes calculated to embed consistent geometric grounding and physics-aware mechanisms. In sum, our results attest at once to remarkable selective gains and to the persistence of fundamental gaps in holistic spatial intelligence.

Table 10: Multiple-choice evaluation accuracy on SPATIALAB-MCQ, disaggregated by question sub-category, for the spatial reasoning models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

E.2 MODEL FAILURE ANALYSIS

The distribution across categories and sub-categories in Table 11 discloses several notable patterns in model performance. Categories such as 3D Geometry and Size & Scale exhibit comparatively balanced distributions, yet both retain a substantial proportion of failures in which no model answered correctly, an indication that even elementary geometric invariances are not reliably internalized. Depth & Occlusion, by contrast, exhibits a heavier tail in the mid-range (2–4 correct), which points to partial but inconsistent reasoning about layered scenes. Orientation emerges as a weak category overall, with especially low totals in Stacking Orientation and Tool Handedness, whence reasoning about subtle directional or embodied affordances is poorly captured. Relative Positioning presents a mixed picture: while certain tasks such as Proximity Gradients register moderate success, others such as Corner/Angle Positioning accumulate many near-zero correct counts, a reflection of brittleness in compositional reasoning over reference frames. Spatial Navigation, intriguingly, exhibits comparatively stronger performance, most notably in Obstacle Avoidance and Viewpoint Visibility, and yet retains substantial errors in Pathway Existence, where relational chaining is requisite. In the aggregate, no category attains uniformly high accuracy, and the Grand Total exhibits an even spread across the 0–5 bins, which intimates that success is highly task-dependent rather than indicative of general spatial competence. On the whole, models display isolated strengths yet want systematic reliability across the spatial dimensions.

These results intimate that contemporary vision–language models labor under several root causes bound up with representational and reasoning bottlenecks. The high error rates in Stacking Orientation and Tool Handedness betoken limited grounding in physics-based and embodied reasoning, consistent with models trained predominantly on static internet imagery rather than on interactive or task-driven data. The difficulty attending Corner/Angle Positioning and Pathway Existence reflects inadequate multi-step relational chaining, and thereby points to weaknesses in the models’ capacity to negotiate recursive spatial logic. The inconsistency observed in Transparency Effects and Reflective Surfaces underscores the challenge of accommodating visual phenomena that hinge upon non-local cues, secondary reflections or occluded geometry among them. The comparatively stronger performance in

Table 11: Distribution of question counts according to the number of models (0–5) that answered each item correctly, disaggregated across categories and sub-categories in SPATIALAB-MCQ. A value of “0” denotes that no model answered correctly, whereas “5” denotes that only five of the models tested succeeded in answering.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

Obstacle Avoidance, by contrast, intimates that models may have acquired shortcuts from dataset biases, such as the frequent co-occurrence patterns of walkable paths, in lieu of genuine planning ability. Considered jointly, these failures accentuate that prevailing training regimes overemphasize correlation-driven visual–language alignments while underrepresenting embodied, counterfactual, and physics-based reasoning. Redressing these deficiencies will in all likelihood require richer training distributions (e.g., embodied simulation data and physics-augmented learning) together with architectural innovations capable of explicitly modeling reference frames and relational constraints. In the absence of such advances, models will continue to exhibit a brittle competence that disintegrates under compositional or physically grounded tasks.

Table 12: Open-ended evaluation accuracy on SPATIALAB-OPEN, disaggregated by question sub-category, for the closed-source models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). We observe that closed-source models attain the strongest overall performance under open-ended evaluation, and yet their results remain uneven across sub-categories. GPT-5-mini stands as the unambiguous leader, consistently securing the best scores in **3D Geometry** and **Relative Positioning**, with peaks of 58.14% in Directional Relations and 55.81% in Alignment Patterns. Gemini-2.5-Flash and Claude 3.5 Haiku, by contrast, register decidedly weak results in Proximity Gradients (16.28%) and Perspective Distortion (12.0%), respectively. This pronounced gap throws into relief the fragility of even the strongest models when charged with reasoning over subtle, continuous visual cues. We attribute these disparities to the differential sophistication of the models’ spatial reasoning pipelines, wherein certain architectures apprehend relational and geometric structure more effectively than others. Tasks such as Proximity Gradients prove especially exacting, inasmuch as they call for analogical, graded reasoning rather than discrete classification. That GPT-5-mini succeeds where others fail suggests that scaling and proprietary training corpora confer distinct advantages. Even the leader, however, betrays vulnerabilities, and thereby points to a systemic issue: closed-source dominance is relative rather than absolute, and these models still fall short of robust, human-like spatial reasoning.

Open-source Large Models (Table 13). For large open-source models, we discern a more dispersed performance landscape, with different systems excelling in isolated sub-categories. llama-3.2-90b-vision-instruct achieves 60.0% in Depth and Occlusion, while GLM-4.5V-106B-MoE leads in Spatial Containment at 38.0%. Notwithstanding these highlights, several models collapse on Proximity Gradients, qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct alike scoring a mere 9.3%. This outcome indicates that architectural scaling assists without guaranteeing coverage across the more demanding continuous-reasoning tasks. We hypothesize that much of this variation issues from heterogeneous training data: models exposed to richer depth and geometry corpora perform better on occlusion and containment tasks, whereas others generalize poorly. The persistent weakness on Proximity Gradients suggests that these systems are not optimized for subtle spatial analogies and distance scaling. We further observe higher inter-model variance within this group

Table 13: Open-ended evaluation accuracy on SPATIALAB-OPEN, disaggregated by question sub-category, for the large open-source models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
Depth and Occlusion	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
Orientation	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
Relative Positioning	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	26.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
Spatial Navigation	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	28.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

than among the closed-source models, an observation that underscores the impact of open-source development heterogeneity. On the whole, large open-source models exhibit encouraging peaks alongside inconsistent coverage, whence scale alone proves insufficient in the absence of targeted pre-training and fine-tuning.

Open-source Small Models (Table 14). Small open-source models predictably trail their larger counterparts, yet their results reveal noteworthy localized strengths. qwen-2.5-vl-3b-instruct achieves 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads in Spatial Containment (24.0%) and Stability Prediction (21.43%). These successes are, however, offset by catastrophic failures: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B manages a mere 2.0% in Shape Projection. We attribute this extreme variability to limited capacity in concert with restricted training diversity, which together preclude small models from acquiring robust representations of geometric and relational principles. Sub-categories that exact precise reasoning about orientation and hierarchy precipitate complete breakdowns, since minor misinterpretations translate into total failure. Their occasional capacity to outperform larger models on narrow tasks, nonetheless, suggests that even small-scale training can instill isolated competencies, a fact that at once attests to the value of specialization and confirms the structural ceiling intrinsic to compact architectures. In short, small open-source models exhibit pockets of promise while remaining highly brittle, their overall accuracy circumscribed by constraints of size and data.

Reasoning Models (Table 15). Reasoning-oriented models present a sharper internal divide, with Gemini-2.5-flash-thinking towering over its peers. It records the best scores in nearly every sub-category, Tool Handedness at 75.0% and Complete Occlusion Inference at 57.14% among them. Kimi-VL-A3B-Thinking-2506, by contrast, struggles dramatically, scoring a scant 6.0% in Spatial Containment and lagging across the board. We attribute this gap to differences in the manner in which these models integrate multi-modal streams into coherent reasoning frameworks. Gemini-2.5-flash-thinking appears capable of forming and interrogating internal world models, a facility that enables it to infer object interactions and physical properties the more effectively. Weaker models, by contrast, lean heavily upon surface-level pattern matching, which proves inadequate to compositional or causal reasoning. The consistently high performance of Gemini-2.5 across categories suggests that explicit reasoning mechanisms yield tangible benefits for spatial understanding. Its failure to dominate Shape Projection and Betweenness Relationships, however, demonstrates that even advanced reasoning modules are not yet universally effective. Thus, although reasoning models hold promise, the category itself remains heterogeneous and far from solved.

Spatial Reasoning Models (Table 16). Spatial reasoning models, notwithstanding their explicit design for the task, record the lowest results overall. Performance is highly fragmented: several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction.

Table 14: Open-ended evaluation accuracy on SPATIALAB-OPEN, disaggregated by question sub-category, for the small open-source models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

Table 15: Open-ended evaluation accuracy on SPATIALAB-OPEN, disaggregated by question sub-category, for the reasoning models. We bold and underline the best score within each model category.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

SpaceQwen2.5-VL-3B-Instruct distinguishes itself marginally, with 23.91% in Volume Comparison and 13.73% in Scale Consistency, yet these scores remain far beneath competitive thresholds. At the opposite extreme, Betweenness Relationships and Corner/Angle Positioning collapse in their entirety, models recording 0.0% and 2.78%, respectively. These results suggest that the models’ diminished capacity, in conjunction with their narrowly focused training, severely constrains their representational flexibility. We construe these failures as evidence that narrow specialization, unaccompanied by sufficient capacity or data breadth, cannot yield generalizable spatial reasoning. On geometric and positional tasks in particular, even slight representational errors precipitate total breakdowns.

Table 16: Open-ended evaluation accuracy on SPATIALAB-OPEN, disaggregated by question sub-category, for the spatial reasoning models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	16.0
	Shape Projection	8.0	8.0	6.0
	Spatial Containment	8.0	10.0	10.0
	Stability Prediction	14.29	14.29	7.14
	Volume Comparison	19.57	23.91	23.91
Depth and Occlusion	Complete Occlusion Inference	7.69	10.26	7.69
	Layering Order	8.57	11.43	4.29
	Partial Occlusion	10.0	12.0	4.0
	Reflective Surfaces	4.0	4.0	2.0
	Transparency Effects	4.0	8.0	2.0
Orientation	Cardinal Direction	18.0	24.0	8.0
	Facing Direction	23.81	23.81	16.67
	Object Rotation	20.0	20.0	18.0
	Stacking Orientation	8.57	8.57	17.14
	Tool Handedness	0.0	4.0	8.0
Relative Positioning	Alignment Patterns	32.56	32.56	23.26
	Betweenness Relationships	4.26	2.13	0.0
	Corner/Angle Positioning	2.78	2.78	2.78
	Directional Relations	11.63	6.98	11.63
	Proximity Gradients	6.98	6.98	9.3
Size and Scale	Distance-Size Correlation	18.0	18.0	16.0
	Perspective Distortion	16.0	20.0	2.0
	Relative Size Comparison	22.0	20.0	8.0
	Scale Consistency	9.8	11.76	13.73
	Shadow-Size Projection	27.45	27.45	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	9.76
	Obstacle Avoidance	10.0	8.0	14.0
	Pathway Existence	6.52	6.52	8.7
	Spatial Sequence	16.0	12.0	16.0
	Viewpoint Visibility	20.0	16.0	8.0

Consequently, notwithstanding their focus, spatial reasoning models prove least effective, an outcome that underscores the necessity of both scale and diversity in the attainment of robust performance.

Overall Discussion. Across all categories, closed-source (Table 12), open-source large (Table 13), open-source small (Table 14), reasoning (Table 15), and spatial reasoning (Table 16), we find a consistent pattern of fragmented competence attended by systemic weaknesses. Closed-source models perform best overall, GPT-5-mini attaining highs of 58.14% in Directional Relations and 55.81% in Alignment Patterns, and yet even they collapse on Proximity Gradients. Large open-source models exhibit impressive peaks, 60.0% in Depth and Occlusion among them, but nonetheless fail at a scant 9.3% in Proximity Gradients, an illustration that scale bereft of specialization remains insufficient. Small open-source models corroborate this further, evincing rare localized strengths (e.g., 26.0% in Relative Size Comparison) alongside catastrophic lows such as 0.0% in Betweenness Relationships. Reasoning models demonstrate that explicit reasoning mechanisms can markedly improve results, Gemini-2.5-flash-thinking reaching 75.0% in Tool Handedness, and yet peers such as Kimi-VL-A3B descend to single digits, thereby exposing the fragility of less-integrated reasoning frameworks. Spatial reasoning models, their specialization notwithstanding, underperform the most, with scores frequently beneath 15% and total failures on Betweenness Relationships. Considered in aggregate, these results implicate three root causes, namely (1) an absence of robust representations for continuous and analogical spatial cues, (2) structural limitations bound up with scale and training diversity, and (3) brittle reasoning pipelines that preclude generalization across tasks. We conclude that no current approach, be it proprietary scaling, open-source expansion, reasoning integration, or specialization, delivers comprehensive coverage. The frequent performance gaps exceeding 50 percentage points within a single model further confirm that progress remains piecemeal. Future work must therefore concentrate upon unifying robust geometric encodings, physics-informed reasoning, and cross-task transfer, so as to advance beyond isolated gains toward consistent spatial intelligence.

Table 17: Distribution of question counts according to the number of models (0–5) that answered each item correctly, disaggregated across categories and sub-categories in SPATIALAB-OPEN. A value of “0” denotes that no model answered correctly, whereas “5” denotes that only five of the models tested succeeded in answering.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

F.2 MODEL FAILURE ANALYSIS

The distribution in Table 17 foregrounds persistent weaknesses across categories, with a pronounced skew toward the lower bins (0–2 models correct). 3D Geometry proves broadly challenging, Shape Projection (19 items with zero correct) and Stability Prediction (10 items with zero correct) exhibiting especially poor outcomes, whence the models want reliable physical and geometric reasoning. Depth and Occlusion emerges as the most arduous category overall, accumulating the highest zero-correct counts (72), most conspicuously in Layering Order (20 zero) and Transparency Effects (16 zero), both of which exact a nuanced treatment of occlusion and non-local cues. Orientation exhibits analogous struggles: whereas Cardinal Direction attains some mid-range success, Stacking Orientation (11 zero) and Tool Handedness (5 zero, with none achieving perfect success) underscore the brittleness of

embodied directional reasoning. Relative Positioning reveals widespread inconsistency, tasks such as Betweenness Relationships and Proximity Gradients being solved by more than two or three models only rarely, while Corner/Angle Positioning exhibits low yet somewhat balanced performance. Size and Scale appears comparatively stronger, several subcategories (e.g., Scale Consistency, with 6 full successes) demonstrating that certain models internalize coarse perspective and scaling cues, though failures remain high in Shadow-Size Projection (17 zero). Spatial Navigation is divided: whereas Spatial Sequence and Obstacle Avoidance register occasional full-model success, Pathway Existence (16 zero) and Accessibility Constraints (10 zero) foreground persistent difficulty in multi-step relational chaining. On the whole, the grand total reveals that, across 356 items, zero-correct predominates (356), while a mere 74 items are solved by all five models, a distribution that confirms the elusiveness of consistent spatial competence.

These findings indicate that the root causes of failure derive less from dataset design than from the representational and reasoning limitations of current models. In the first instance, the preponderance of errors in Depth and Occlusion reflects a reliance upon superficial appearance features in preference to structured scene representations: models may fasten upon surface textures rather than reason about hidden layers, an inclination that engenders systematic errors in Layering Order. Failures on Orientation tasks such as Tool Handedness expose the absence of embodied affordance modeling, inasmuch as pretrained internet imagery frequently lacks explicit annotations of left/right handedness or stacking physics. The brittleness observed in Relative Positioning and Spatial Navigation points to deficiencies in the chaining of multi-hop spatial relations; Pathway Existence, for instance, requires the integration of local connectivity with global layout, which attention-based co-occurrence priors capture poorly. The partial success attained in Size and Scale, by contrast, intimates that models exploit statistical regularities in perspective distortion or scale consistency, only to collapse under less frequent cases such as shadows. These patterns collectively demonstrate that current models succeed wherever cues are salient and correlate with training priors, yet fail wherever reasoning demands counterfactual, compositional, or embodied inference. Redressing these deficiencies will require architectural advances that explicitly model reference frames, occlusion layers, and physical dynamics, together with training data enriched by embodied simulations and counterfactual augmentations. In the absence of these, progress in spatial reasoning will remain fragmented, with models performing well on surface-level cues yet disintegrating under structured, real-world spatial logic.

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

In the present section we furnish a detailed analysis of the performance gap that separates the multiple-choice (MCQ) and open-ended evaluation formats across the spatial reasoning categories of SPATIALAB, as documented in Table 18.

G.1 QUANTITATIVE SNAPSHOT

Averaged across 25 models, the performance gap separating the MCQ and open-ended formats amounts to **23.0%**, with a standard deviation of 5.5%. The dataset-wide means per subtask are as follows: 3D geometry (24.57), depth & occlusion (23.45), orientation (23.70), relative positioning (23.11), size & scale (23.42), and spatial navigation (22.89), yielding an overall average of 23.01. The overall gap ranges from a minimum of 11.93 to a maximum of 31.57 (median $\approx$ 23.34). At the family level, spatial reasoning models manifest the widest average gap (27.03), whereas reasoning-oriented models exhibit the narrowest (19.11); proprietary models incur lower gaps on average (21.09) than many long open-source models (24.51). Notably, the human baseline registers an overall gap of 22.64. The widest gaps obtain for InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%), while the narrowest accrue to Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36). Correlational analysis reveals that spatial navigation predominates in the overall gap (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) likewise exhibiting strong alignment.

G.2 PATTERNS AND OBSERVATIONS

Three broad patterns emerge. First, the specialist spatial reasoning models exhibit the widest MCQ→open-ended gaps, an indication that they are optimized for classification-style supervision

Table 18: **Performance gap between the MCQ and Open-ended formats (%) ($\downarrow$) across the spatial categories of SPATIALAB.** A lower value signifies that a model performs comparably under both setups, whereas a higher value bespeaks greater disparity. We bold and underline the widest gap within each model group.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
Open-Source Models (Small)							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
Reasoning Models							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
Spatial Reasoning Models							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

and correspondingly less robust to free-form generation; reasoning-focused models, by contrast, display narrower gaps, a reflection of the stabilizing effect that chain-of-thought-style training exerts upon generative outputs. Second, spatial navigation constitutes the strongest predictor of overall disparity, in that models which falter on open-ended navigation reasoning drive the aggregate gap; orientation and 3D geometry subtasks contribute substantially as well, whereas relative position and size/scale vary more idiosyncratically across models. Third, scale alone does not account for robustness: smaller models such as Llama-3.2-11B attain among the narrowest gaps, while very large models such as InternVL3.5-72B exhibit the widest. Finally, several negative or near-zero gaps surface (e.g., depth & occlusion for Llama-3.2-11B at -1.98 , and relative position for o4-mini at -3.18), an indication that in certain cases open-ended evaluation proves more reliable than MCQ, in all likelihood by reason of misleading distractors.

G.3 HYPOTHESES ON ROOT CAUSES

We hypothesize several interacting factors that underlie the observed MCQ $\rightarrow$ open-ended performance gaps in SPATIALAB, each of which is corroborated by statistical observation:

1. **MCQ structural advantage.** The MCQ format constrains the output space, thereby permitting models to exploit surface cues or to eliminate distractors. Across models, the mean MCQ score is systematically higher than the open-ended score, the average gap standing at 23.0% with $\sigma = 5.5\%$. The negative gaps observed for Llama-3.2-11B in depth & occlusion (-1.98) and for o4-mini in relative position (-3.18) indicate that open-ended generation can, on occasion, outperform MCQ wherever distractors prove misleading, a finding that reinforces the notion that MCQ is liable to inflate or distort measured performance artificially. *Recommendation:* To address this issue, we suggest complementing MCQ evaluations with open-ended tasks and subtask-level diagnostics, which can more accurately capture model competence and reduce overestimation caused by option-based cues.

2. **Specialization bias in spatial reasoning models.** Spatial reasoning models exhibit the widest average gaps (27.03%) notwithstanding their optimization for spatial tasks, an indication that they may have been trained predominantly on categorical or synthetic selection tasks rather than on free-form generation. InternVL3.5-72B, a 72B-parameter spatial reasoning model, for instance, exhibits the widest overall gap (31.57%), most conspicuously in spatial navigation (36.68%) and orientation (34.44%), a pattern that foregrounds a reliance upon classification-like supervision. The correlations between the subtask gaps and the overall gap (SpNav: $r = 0.99$, Orientation: $r = 0.83$) indicate that these format-dependent failures are concentrated in sequential and geometric reasoning tasks.

Recommendation: To mitigate this bias, we recommend fine-tuning or instruction-tuning on open-ended spatial generation tasks, which can improve model robustness to generative formats and reduce reliance on categorical supervision.

3. **Instruction-tuning and stepwise decoding.** Reasoning-oriented models (e.g., Gemini-2-Flash-Thinking, o4-mini) demonstrate narrower average gaps (around 19.11%) attended by lower variance across subtasks. Gemini-2-Flash-Thinking, for example, exhibits an overall gap of 13.0% together with consistently modest subtask gaps (Difference in 3D Geometry: 13.9%, Difference Relative Positioning: 21.03%), an indication that chain-of-thought or stepwise decoding assists models in producing stable open-ended responses. This inference is further buttressed by the reduced correlation between subtask-gap variance and overall gap within this family, which implies that instruction-tuned reasoning attenuates format sensitivity.

Recommendation: To address format sensitivity, we suggest adopting chain-of-thought or stepwise decoding strategies and instruction-tuning for open-ended spatial tasks, which can improve generative consistency and reduce MCQ dependence.

4. **Spatio-navigation stresses sequential grounding.** The spatial navigation gaps constitute the strongest predictor of overall MCQ→open-ended disparity (Pearson $r = 0.99$). Models beset by high performance gaps in spatial navigation, such as SpaceQwen2.5-VL-3B-Instruct (Spatial Navigation gap = 35.87%), likewise exhibit wide overall gaps (29.78%), an indication that multi-step referential reasoning is disproportionately impaired in open-ended formats. This suggests that MCQ options simplify the reasoning chain, thereby masking underlying weaknesses in sequential grounding and relational computation.

Recommendation: To better capture sequential reasoning abilities, we recommend including dedicated open-ended spatio-navigation tasks and employing targeted multi-step reasoning supervision, which can improve grounding and reduce overestimation by MCQ.

5. **MCQ distractor artifacts.** Certain negative or near-zero gaps indicate that poorly designed distractors can suppress measured MCQ performance. o4-mini, for example, exhibits a negative relative-position gap (−3.18%) while sustaining positive gaps on other subtasks, which implies that distractor quality can artificially depress MCQ accuracy. Conversely, models such as Qwen-VL2.5-32B-Instruct exhibit very wide MCQ→open gaps (overall = 29.85%) across multiple subtasks, an indication that MCQ distractors can also exaggerate perceived competence in selection-based tasks.

Recommendation: To address this, we suggest careful auditing and refinement of MCQ distractors and complementing them with open-ended evaluation, which can yield more accurate and unbiased measures of model spatial reasoning ability.

G.4 PRACTICAL IMPLICATIONS

Our findings evince the importance of reporting both MCQ and open-ended results. Exclusive reliance upon MCQ inflates estimates of practical spatial competence, the average disparities hovering around 23%. Subtask-level breakdowns prove indispensable, inasmuch as spatial navigation disproportionately drives the observed differences. Diagnostic evaluations that disentangle classification from generative reasoning can the more effectively isolate error modes. Instruction-tuning and decoding strategies conducive to stepwise reasoning, moreover, appear promising as means of reducing MCQ dependence. Finally, the careful design and auditing of MCQ distractors are requisite if fair comparisons across formats are to be secured.

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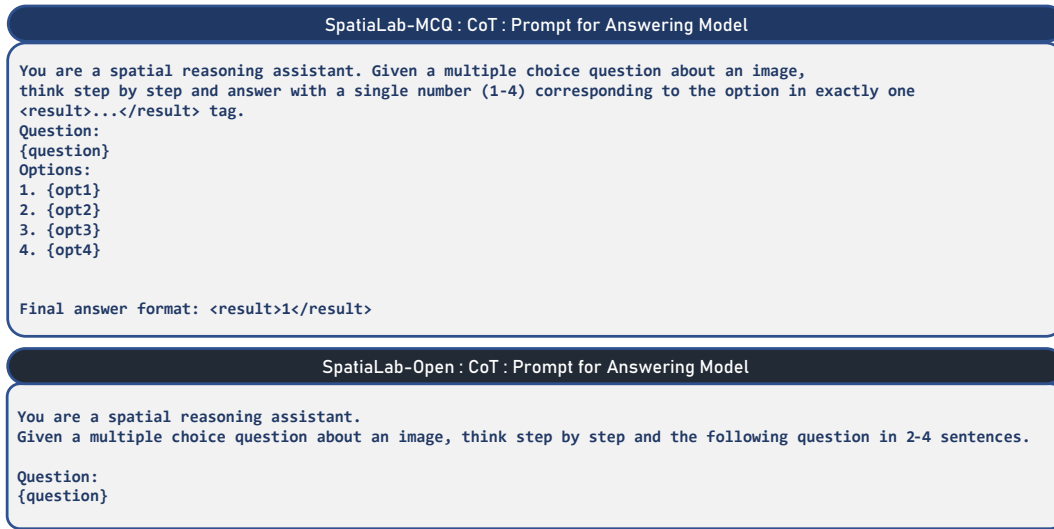


Figure 7: The Chain-of-thought (CoT) prompts.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

With a view to enhancing spatial reasoning performance on our diverse visual question benchmark, we explored a plurality of complementary strategies.

H.1 INHERENT REASONING MECHANISMS

We next scrutinize the role of built-in reasoning mechanisms by juxtaposing standard models with their reasoning-enabled counterparts (Table 2, Table 3). Across the MCQ evaluations, reasoning consistently augments performance, albeit to a degree that varies by category. Gemini-2.5-Flash-Thinking, for instance, improves upon its non-reasoning variant by +4.64% overall (52.93% vs. 48.29%), with particularly pronounced gains in Relative Positioning (+13.1%) and Orientation (+8.6%). An analogous trend obtains for o4-mini-medium, which attains the highest overall MCQ accuracy (53.21%), an outcome suggestive of the particular efficacy of explicit reasoning scaffolds within structured, discrete-answer settings.

Open-ended evaluation, by contrast, discloses a more nuanced picture. Here again the reasoning-equipped models surpass their baselines, though the gains prove uneven. Gemini-2.5-Flash-Thinking improves marginally upon Gemini-2.5-Flash (32.77% vs. 30.93%), yet the most substantial leap is observed in o4-mini-medium (37.86%), which markedly outstrips all non-reasoning models. These results indicate that reasoning mechanisms help to mitigate the instability characteristic of free-form generation, most conspicuously in complex categories such as 3D Geometry and Spatial Navigation. The gains, however, are not universal: reasoning models occasionally exhibit regressions, as when Gemini-2.5-Flash-Thinking declines sharply in Size & Scale (21.74% vs. 26.59%), thereby echoing our earlier finding that generative settings amplify brittleness.

On the whole, these comparisons accentuate that explicit reasoning modules substantially enhance structured MCQ performance and can stabilize open-ended responses, and yet their benefits accrue unevenly across categories. The results intimate that reasoning mechanisms fortify logical consistency and relational chaining while failing to resolve in full the challenges posed by scale sensitivity, occlusion, and free-form linguistic grounding.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

In order to ascertain whether structured reasoning improves spatial performance, we further evaluated models under Chain-of-Thought (CoT) prompting. For each of the 30 sub-categories in SPATIALAB, we sampled 5 items (with the seed fixed to 42) so as to constitute a balanced test set spanning all major spatial phenomena, a design that secures equitable coverage while rendering the evaluation

computationally tractable. Exemplary CoT prompts are exhibited in Figure 7, which illustrate the step-by-step reasoning scaffolds furnished to the models. This setup enables us to isolate the contribution of CoT reasoning from that of model scale or training.

Table 19: Performance under CoT prompting across categories for InternVL-3-78b and Gemini-2.5-Flash. Values denote accuracies (%), and Gains represent the difference between the settings with and without CoT prompting.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

Our analysis (Table 19) demonstrates that orientation is the sole category in which CoT prompting reliably improves performance, while most other categories undergo either stagnation or decline. This pattern intimates that reasoning avails wherever the task is reducible to the logical alignment of directions, yet founders wherever more complex perceptual priors (e.g., depth, scale, relative positioning) are requisite. We surmise that this obtains because models want inherent knowledge of 3D perception and geospatial relationships; they are not, at bottom, “aware” of these concepts in the manner of humans. Consequently, once CoT prompting is applied, the step-by-step reasoning does not rectify misconceptions but rather entrenches them, propagating and amplifying errors across the reasoning steps.

This finding accentuates the non-triviality of our benchmark: it cannot be solved merely by grafting multi-step reasoning onto existing models. In contradistinction to textual reasoning tasks, wherein CoT reliably improves outcomes, our benchmark lays bare deeper architectural limitations in perceptual and spatial grounding. In categories such as depth and size, incorrect priors ensure that each additional reasoning step displaces the model further from the correct solution. The sharp performance declines, most conspicuously for Gemini in MCQ settings, illustrate how fragile reasoning becomes once it is unsupported by robust perceptual knowledge.

Our task design thus constitutes a strength, in that it transcends surface-level reasoning and probes whether models possess a genuine understanding of geospatial and 3D relational concepts. To solve it requires more than the chaining of logic; it demands embeddings and training that capture core facets of perception and spatial cognition. In this respect, our benchmark challenges models at a fundamental level and throws into relief an important gap in current architectures.

H.3 CHAIN-OF-THOUGHT (CoT) PROMPTING WITH SELF-REFLECTION

We further extend the CoT setup by interposing a self-reflection stage, wherein the model is prompted to review and revise its initial reasoning prior to finalizing an answer (see Figure 8). The evaluation protocol remains identical to that of the standard CoT setting, drawing upon 5 balanced samples per sub-category (seed 42). This design isolates the effect of self-reflection while keeping the comparison directly aligned with baseline CoT performance.

Table 20: Performance under CoT versus CoT augmented with Self-Reflection (CoT-SR) across categories for InternVL-3-78b and Gemini-2.5-Flash. Values denote accuracies (%), and Gains represent the difference between CoT-SR and CoT.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67



Figure 8: The Chain-of-thought (CoT) with Self-Reflection prompts.

The augmentation of Chain-of-Thought (CoT) prompting with self-reflection produced mixed and markedly model-dependent outcomes. For InternVL-3-78b, overall accuracy declined slightly (−2.66% MCQ, −8.67% open-ended), with sharp reductions in categories such as depth, relative positioning, and navigation, an indication that reflection frequently entrenched errors rather than correcting them. Certain localized gains were observed, +12% in Size & Scale (MCQ) among them, yet these proved inconsistent and were offset by substantial degradations (−28% in relative positioning, open-ended). Gemini-2.5-Flash, by contrast, exhibited strong MCQ gains (+20.66%), with particularly pronounced improvements in 3D Geometry (+32%) and Depth & Occlusion (+24%), though its open-ended improvements remained negligible (+0.67%). These results foreground a key asymmetry: reflection avails in multiple-choice contexts, wherein models may prune improbable options, yet fails to transfer to free-form generation, wherein uncertainty propagates into ungrounded rationales. We further find that the categories most improved by reflection are precisely those in which geometric consistency is central (orientation, depth, size), whereas tasks demanding relational integration across multiple objects (relative positioning, navigation) deteriorate. On the whole, reflection operates as a stabilizer within structured choice settings while failing to enhance naturalistic reasoning reliably.

This observed inconsistency mirrors our earlier findings with plain CoT prompting: reasoning enhancements succeed only where the model possesses stable perceptual anchors upon which to ground its steps. Self-reflection interposes an explicit verification pass, and yet this mechanism is only as robust as the model’s internal priors. Wherever depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass amplifies misconceptions rather than repairing them. This accounts for the fact that Gemini, which exhibits stronger perceptual grounding, benefits under MCQ

conditions, whereas InternVL-3-78b sustains more frequent degradations. The broader root cause resides in a pretraining mismatch: current VLMs import reasoning templates from language modeling yet want the embedded 3D and geospatial representations upon which humans draw. In the absence of such anchors, reflection cannot effect true error correction; it functions instead as a linguistic filter that burnishes reasoning syntax without improving perceptual grounding. As a consequence, reflective reasoning loops afford modest utility within discrete answer formats but collapse under open-ended demands, wherein factual verification is wanting.

These findings confirm that our benchmark exposes a fundamental gap in current architectures: surface-level reasoning layers alone cannot compensate for absent spatial cognition, and meaningful progress will require a deeper integration of geometric and perceptual structure. Moreover, since VLLMs are commonly trained on web data, the concepts well represented online are precisely those on which models tend to perform better, whereas poorly represented spatial and 3D concepts fare worse, a circumstance that reinforces these performance disparities.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We fine-tuned the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) by means of a parameter-efficient approach predicated upon LoRA (Hu et al., 2021). In order to accommodate constrained GPU resources, we loaded the model in 4-bit precision (`load_in_4bit=True`) and inserted LoRA adapters into the attention and feed-forward network projection layers. The LoRA hyperparameters, rank, scaling factor, dropout, and target modules among them, are recapitulated in Table 21. Gradient checkpointing was enabled so as to curtail memory usage, and a fixed random seed secured reproducibility.

For tokenization, we adhered to the Qwen-2.5 chat template in order to preserve consistent formatting between training and inference. Each training example was converted into a single text sequence, and dynamic padding was applied by way of `DataCollatorForSeq2Seq`. Training proceeded at an effective batch size of 4 (a per-device batch size of 1 in concert with gradient accumulation of 4). The trainer and optimization hyperparameters, learning rate, optimizer, weight decay, and scheduler among them, are enumerated in Table 22. The training loop comprised four passes, each spanning a single epoch, and thereby yielded four saved checkpoints.

On the whole, this setup enabled us to fine-tune a large vision–language model efficiently under constrained GPU resources while preserving a high adaptation capacity.

Table 21: LoRA hyperparameters employed for SFT.

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trends in Figure 4a reveal notable divergences between the MCQ and Open-ended evaluation modes over the course of the training epochs. We observe that MCQ accuracy commences from a lower baseline of 25.63% yet rises consistently, converging near 36.59% by the fourth epoch. This steady improvement intimates that the model derives substantial benefit from additional supervised fine-tuning, most notably on tasks that confine answers within a limited candidate space. Open-ended evaluation, by contrast, begins higher, at 34.40%, only to decline sharply by the second epoch, plummeting to as low as 12.62%; not until the later epochs does it partially recover, reaching 35.48% at the close. This instability betokens a sensitivity intrinsic to open-ended generation that is absent from the MCQ setting, wherein the structure of multiple-choice questions affords more

Table 22: Trainer and optimization hyperparameters employed for SFT.

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

stable optimization signals. We construe the fluctuations in open-ended performance as evidence that fine-tuning may overfit to MCQ-like reasoning patterns at the expense of free-form linguistic generalization. The dotted baselines in the plot further accentuate how the MCQ mode achieves gains beyond its initial standing, whereas the Open-ended mode remains approximately stagnant. On the whole, the figure illustrates that SFT preferentially favors MCQ tasks with respect both to stability and to consistent accuracy gains.

The tabulated results in Table 4b complement the figure by supplying a detailed before-and-after breakdown across the six spatial reasoning categories. On the MCQ tasks, we find that performance improved in every category, the gains ranging from a modest 1.59% in Relative Positioning to a substantial 17.11% in Size and Scale; the Grand Total gain for MCQ stands at 10.97%, a figure that confirms the efficacy of the fine-tuning procedure in enhancing structured-answer performance. The Open-ended results, by contrast, prove considerably more mixed, encompassing both positive and negative changes. Categories such as 3D Geometry and Spatial Navigation exhibit small but positive gains (3.50% and 2.82%, respectively), and yet Size and Scale registers a pronounced decline of -3.95% , while Relative Positioning likewise decreases by -0.79% , a pattern that reflects the capacity of fine-tuning inadvertently to impair certain reasoning modes in free-form contexts. The overall Open-ended gain amounts to a mere 1.07%, statistically negligible by comparison with the MCQ improvements. These results confirm that the efficacy of SFT is highly sensitive to the evaluation paradigm; by isolating the categories, we bring into relief where training avails, where it fails, and how the performance benefits differ between constrained and unconstrained outputs.

A more penetrating inspection reveals striking disparities between categories and formats. On the MCQ tasks, Size and Scale together with Spatial Navigation yield the highest improvements, an indication that models trained with multiple-choice prompts prove especially adept at learning tasks that demand scale discrimination and navigational reasoning. In Open-ended form, however, Size and Scale in fact declines, a demonstration that the same learned representations transfer poorly to generative contexts. In like manner, Relative Positioning scarcely improves under MCQ and deteriorates under Open-ended, thereby foregrounding a consistent weakness in relational reasoning irrespective of format. Orientation, conversely, exhibits strong MCQ gains ($+13.33\%$) alongside small Open-ended improvements, an indication that certain categories generalize the more robustly. We hypothesize that the divergence between the two formats may issue from the structural scaffolding that MCQs provide, which diminishes the necessity for the model to generate precise linguistic formulations. Open-ended tasks, by contrast, exact simultaneous mastery of reasoning and natural-language fluency, which amplifies pre-existing weaknesses. The trends indicate that, although fine-tuning strongly enhances pattern recognition within discrete-choice settings, it introduces volatility and even degradation wherever free expression is required. This duality makes plain that training interventions optimized for MCQs may not generalize seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, we demonstrate that SFT, as presently applied, confers strong benefits upon structured tasks yet affords limited or even negative transfer to open-

ended reasoning. This limitation is not merely an artifact of fine-tuning but is, moreover, symptomatic of the manner in which contemporary VLMs are pretrained. Existing pretraining pipelines emphasize heavily broad image–text alignment, object recognition, and captioning, and yet furnish little explicit supervision as regards compositional spatial structures, occlusion layers, or geometric transformations. Models consequently learn to excel at surface-level associations while failing to acquire robust representations of depth, orientation, or scale. The instability we observe in open-ended spatial reasoning intimates that the model’s learned priors are fragile: bereft of multiple-choice scaffolding, they revert to shallow pattern-matching in lieu of grounded inference. Categories such as Size and Scale and Relative Positioning, wherein the declines are most pronounced, point to inductive biases absent from pretraining, biases that humans acquire through embodied interaction with 3D environments. A further contributing factor is the reliance upon synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions, thereby precluding models from generalizing to real-world clutter, noise, and complexity. These findings underscore the need to reconceive both pretraining and fine-tuning for spatial reasoning, incorporating datasets and objectives that reflect naturalistic visual–spatial challenges. Ultimately, we contend that, in the absence of targeted reforms, current VLMs risk entrenching an algorithmic straitjacket: they appear competent under constrained evaluations yet remain brittle and unaligned with the embodied, flexible reasoning that humans bring to bear upon real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

In order to interrogate the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) using the Qwen-2.5-VL-3B-Instruct model. The training trajectory, recapitulated in Table 23, reveals a distinct divergence in the performance dynamics of the Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. As illustrated in Table 23 and Figure 9, performance on SPATIALAB-MCQ exhibits a monotonic improvement, ascending from a baseline of 25.63% to an average of 35.74% at Epoch 4. This trajectory indicates that the model adapts successfully to the discriminative character of the task, learning to map visual features onto constrained output options with escalating reliability.

In sharp contradistinction, SPATIALAB-OPEN displays a pronounced *U-shaped* trajectory across all seeds. In lieu of steady improvement, the model sustains an immediate and significant performance collapse in the early epochs (descending from 34.40% to an average of 19.26% at Epoch 1). Although performance recovers in the later epochs (converging to $\approx 34.82\%$ at Epoch 4), the final gain over the zero-shot baseline is negligible ($+0.42\%$).

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). Whereas MCQ performance improves steadily, Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, an indication of representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

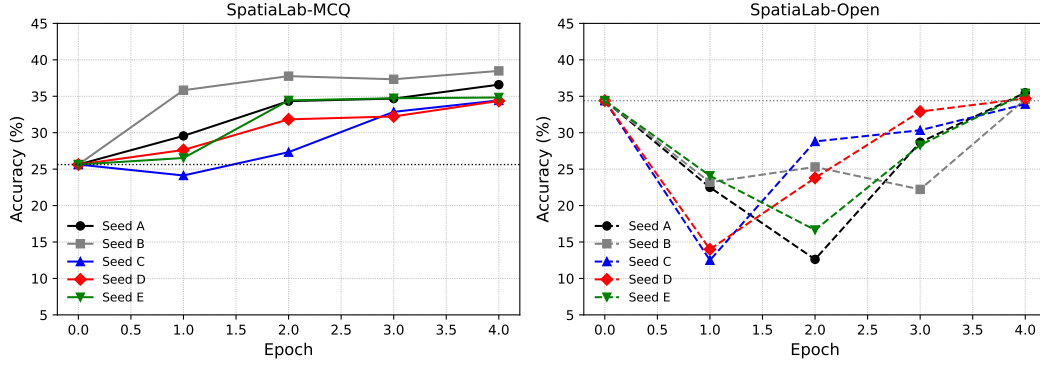


Figure 9: Analysis of the SFT dynamics.

Representational Brittleness and Alignment Tax. We ascribe this phenomenon to the fragility of the spatial representations resident in current VLMs when they are subjected to fine-tuning. The initial collapse in open-ended accuracy is symptomatic of a *catastrophic forgetting* of the linguistic priors requisite for fluent, grounded generation. As the model optimizes for the particular instruction-following format of the training data, it overfits to the structural logic of the tasks (to the benefit of MCQ) at the expense of the flexible, generative spatial grounding that open-ended descriptions require.

The subsequent recovery phase corresponds to the model’s re-learning to articulate spatial concepts within the new distribution, and yet it fails to surpass the initial baseline to any significant degree. This dynamic underscores a critical finding: SFT proves effective in teaching models to *select* correct spatial answers, but it does not fundamentally enhance the underlying *generative spatial representation* requisite for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

In order to ascertain whether the representations acquired from SPATIALLAB generalize beyond the benchmark’s particular format, we evaluated the transferability of our fine-tuned model to three external spatial reasoning datasets, namely OMNISpatial, SPACE (Multimodal), and MIND THE GAP. We employed the Qwen2.5-VL-3B-Instruct model, fine-tuned on SPATIALLAB by way of the SFT protocol detailed in Appendix H.4.1, and evaluated it on these external benchmarks in a zero-shot setting.

Table 24: Transfer-learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks subsequent to fine-tuning on SPATIALLAB. The consistent gains indicate that the model acquires generalizable spatial representations rather than merely overfitting to the format of the source dataset.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As Table 24 attests, fine-tuning on SPATIALLAB yields substantial and consistent performance improvements across all evaluated domains, ranging from +5.24% to +11.70%. The most significant gain was observed on MIND THE GAP (+11.70%), an outcome suggestive of the efficacy with which our benchmark’s emphasis upon occlusion and spatial continuity addresses the reasoning gaps that that dataset targets. In like manner, the improvements on OMNISpatial (+7.05%) and SPACE (+5.24%) confirm that the geometric and relational concepts encoded within our taxonomy accord with broader definitions of spatial intelligence in the literature. These results validate the quality of the SPATIALLAB data. Although the primary purpose of this work is to serve as a rigorous diagnostic benchmark, the strong transferability demonstrates that the dataset does not merely conduce to overfitting upon a specific question template. On the contrary, the diversity of the 30 sub-categories

compels the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), in order to perform structured spatial reasoning over images within the SPATIOLAB evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each apportioned to a dedicated agent, and thereby admits of focused processing of objects, attributes, relations, symmetries, and transformations. This modular design not only augments accuracy and interpretability but further underwrites multi-step reasoning, enabling the system to propagate low-level perceptual cues through higher-order relational and transformational inferences. The architecture, the agent implementation, and their coordinated interaction are detailed in the sections that follow, with the full complement of prompts provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. The Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—geometric shapes, natural elements, and artificial icons among them—alongside attributes such as size, orientation, color, and shading. Relative positions such as leftmost, center, and top are encoded explicitly, and the prompt foregrounds exhaustive detail to the exclusion of summarization. Images are converted into data URIs and dispatched together with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list by way of regex-based extraction. This structured description serves as the foundation for the downstream agents, furnishing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach ensures that even subtle visual features, partial occlusions or minor shape variations among them, are retained for subsequent relational and transformational reasoning.

Object Segmentation Agent. The Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, discriminating among visually similar instances. Each object is assigned a stable identifier (e.g., Obj1, Obj2) so as to sustain consistent references across the subsequent reasoning steps. The agent employs carefully formatted prompts and invokes the same Gemini-2.5 model to structure the description, whereupon post-processing partitions the lines by object ID in order to enforce consistency. This structured representation attenuates ambiguity in the downstream extraction of attributes and relations, and thereby enables the precise mapping of complex scenes. By isolating objects explicitly, the agent further facilitates the detection of overlaps, repetitions, and hierarchical groupings that may not be evident within raw descriptions.

Attribute Extraction Agent. The Attribute Extraction agent translates each object into a structured JSON representation that captures core properties such as shape, size, color, shading, orientation, and positional descriptors. The prompts enforce strict JSON output, and responses are parsed by means of `json.loads()`, with fallback defaults introduced to ensure robustness against model inconsistencies. This structured encoding admits of both quantitative and qualitative comparisons among objects, and thereby underpins the later stages of spatial reasoning. By preserving both absolute and relative measures, the agent maintains fidelity to the original perceptual content while concurrently normalizing descriptors for computational consistency. The JSON structure enables programmatic downstream operations, the calculation of symmetries or the detection of transformations among them.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA, Relation, ObjectB), capturing directional relations (e.g., left_of, above, inside) together with higher-order alignments such as rows, grids, or repetition patterns. Consistent object IDs are enforced so as to forestall relational ambiguity, and parsing partitions the output lines that commence with parentheses, whereupon automated consistency checks are applied. This step transmutes static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure admits of reasoning over both pairwise interactions and more complex scene configurations, a capacity critical to tasks such as symmetry detection, grouping, and the multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. The Grouping and Symmetry agent abstracts higher-order patterns from the object lists and spatial relations, identifying clusters, rows, grids, and symmetries—vertical, horizontal, rotational, and translational alike. Outputs are solicited in structured JSON under the keys `groups` and `symmetries`, with empty structures serving as fallbacks so as to preserve downstream compatibility. This agent captures emergent scene regularities that are not apparent at the level of the object or of the pairwise relation, and thereby furnishes critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent underpins both perceptual inference and anticipatory reasoning about dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes, translations, scaling, rotations, color/shading shifts, and appearance or disappearance events among them. Structured logs are extracted line by line by means of regex patterns that capture directional arrows, and thereby yield a temporal representation of the scene. This encoding enables the system to track not merely static spatial configurations but also dynamic evolution, and thus underpins reasoning about motion, causality, and object interactions over time. The agent ensures that transformations are bound to consistent object IDs, thereby enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates all outputs into a unified JSON format comprising `objects`, `relations`, `groups`, `symmetries`, and `transformations`. The prompts enforce consistent object IDs and structured formatting, while failures of JSON parsing revert to empty dictionaries in order to ensure the robustness of the pipeline. By centralizing all perceptual, relational, and temporal information, this agent constructs a comprehensive scene representation that is at once human-interpretable and machine-readable. The resulting unified structure serves as a versatile foundation for downstream tasks, reasoning evaluation, spatial pattern analysis, and multi-frame prediction among them.

Open-Ended Spatial Reasoning Agent. The standardized representation is fed into the reasoning agent, which answers multiple-choice or pattern-recognition questions. Its inputs comprise objects, attributes, relations, groups, symmetries, and transformations. The model weighs geometric patterns, relational alignment, and dynamic transformations in order to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

For our technical implementation, we employ Gemini-2.5-Flash at low temperature (0.1) so as to secure deterministic outputs, and we perform multiple iterations for refinement. API calls are handled by way of OpenRouter, which supports both text and image inputs, while the parsing of model responses relies upon regex- and JSON-based extraction to generate structured outputs. All interactions are logged in JSON format for the sake of reproducibility, and the final output is a unified JSON per image containing comprehensive information on objects, relations, symmetries, and transformations. This multi-agent decomposition permits each specialized model to concentrate upon a constrained sub-task, and thereby enhances the overall accuracy, interpretability, and robustness of structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol remains identical to that of the standard CoT and CoT-SR setting, drawing upon 5 balanced samples per sub-category (seed 42), for a total of 150 samples.

The evaluation of Gemini-2.5-Flash equipped with agents brings to the fore both the strengths and the limitations of agentic reasoning on perceptual–geometric tasks. Orientation is the sole category in which agents confer consistent and substantial gains, improving by +8.00% in MCQ and by a striking +36.00% in open evaluation. This pattern intimates that agent-based multi-step reasoning proves effective wherever the task admits of decomposition into the sequential alignment of directional cues, wherein structured deliberation offers a manifest advantage over single-step outputs.

Most other categories, by contrast, exhibit negative or neutral gains, an outcome that reveals the non-universality of agent-induced enhancement. Depth & occlusion, for example, sustains a severe −24.00% decline in open evaluation, while spatial navigation decreases by −12.00%. These tasks exact deeper perceptual grounding in three-dimensional space, and the absence of the requisite priors

Table 25: Analysis of agent performance.

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67%	57.33%	+2.66%	35.83%	31.33%	-4.50%

entails that the reasoning steps tend to entrench misconceptions rather than to correct them. In like manner, relative positioning exhibits stagnation under MCQ and a -14.00% decline under open evaluation, an indication that agent loops fail to confer robustness wherever relational understanding is inherently weak.

Certain categories exhibit marginal improvements, size & scale ($+4.00\%$ in MCQ) among them, and yet the gains prove unstable and fail to transfer to open evaluation. Even in 3D geometry the agents fail to furnish a boost, incurring declines of -4.00% and -12.00% in MCQ and open evaluation, respectively. These results underscore that our benchmark is complex and not soluble merely by superimposing multi-step agentic reasoning upon existing models; it demands, rather, a deeper integration of geospatial understanding and perceptual representations, which current architectures want.

On the whole, the mixed results emphasize that agentic workflows are, in themselves, insufficient to advance performance within this domain. Whereas orientation tasks benefit from structured reasoning, the majority of categories lay bare the models' want of inherent awareness of perceptual concepts, a deficiency that causes errors to propagate through the multi-step reasoning and thereby amplifies the performance declines. Consequently, our benchmark functions as a stringent stress test, one that accentuates that future progress will require the embedding of genuine perceptual and geospatial priors rather than a sole reliance upon agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

The present section sets forth a focused qualitative analysis of the model failures observed in our visual reasoning benchmark. We expound each error class, provide a diagnostic root-cause analysis, identify recurring patterns, and describe verification protocols. The discussion aspires to be at once interpretable and actionable, such that it may inform subsequent architectural and evaluation design.

I.1 OVERVIEW

Across the sample set, models attain strong coarse perception, in that they reliably recognize object categories, colors, and simple binary relations, and yet they break down systematically wherever multiple cues must be integrated. The failures constitute not random noise but rather a small set of interpretable classes, namely spatial mislocalization, perspective and scale mistakes, occlusion and ordering errors, attribute confusion, and ungrounded open-ended rationalization. These classes recur across architectures and prompting strategies, a regularity that intimates that the underlying causes are common inductive biases endemic to current vision-language pipelines rather than isolated training artifacts. Significantly, the same error frequently surfaces alike in large-scale proprietary systems and in small open-source baselines, an indication of structural weaknesses shared across the model family. In what follows, we unpack each error class by way of representative examples and root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** Wherever queries require the discrimination of one object from a set of visually similar candidates, models frequently misidentify the target. Asked to indicate “the leftmost red cube among three stacked cubes,” for example, the model may erroneously output the middle cube, swayed by lighting highlights or by central positioning. This difficulty grows particularly acute within cluttered environments such as kitchen counters or crowded table scenes, wherein multiple overlapping objects share the same color and shape. In such cases the model’s errors are not random: it inclines to privilege the objects that appear most salient by reason of size, brightness, or centrality, even where these cues contravene the explicit spatial instructions of the query. These consistent patterns accentuate that the failures are systematic rather than accidental.

★ **Analysis.** The recurring mistakes intimate that models want a robust object-centric representation, or what cognitive science would term an “index” that persists across reasoning steps. In lieu of maintaining a stable pointer to the correct candidate throughout the reasoning chain, models appear to drift, frequently reassigning attention to whichever object seems most visually striking at the intermediate layers. This engenders reasoning errors such as commencing with the leftmost cube yet terminating the chain with the brightest or largest cube. The incapacity to anchor spatial reference consistently renders tasks involving relative comparisons, “the ball behind the second chair” among them, especially fragile. In essence, the model operates more as a salience detector than as a referential reasoner.

⊗ **Root cause.** At the architectural level, feature-pooling operations, whether effected by attention or by convolution, tend to intermix contextual information drawn from neighboring regions. Although this proves beneficial for global context, it effaces the fine-grained, object-specific identity requisite for distinguishing between similar candidates. Once pooled, the features of two adjacent red cubes may prove nearly indistinguishable within the latent space. In the absence of an explicit mechanism, slot-based representations or pointer tokens among them, the network collapses distinct entities into blended embeddings. This structural limitation precludes reliable referent tracking across reasoning steps and accounts for the models’ consistent confusion of dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks that entail metric reasoning about relative size, fit, or scale are handled inconsistently. On simple comparisons such as “is a car larger than a bicycle,” models typically succeed, inasmuch as such relationships accord with strong statistical priors. Confronted with near-threshold decisions such as “will the small sphere fit inside the slightly larger cylinder,” however, errors grow frequent and unpredictable. A common pattern is the reliance upon canonical object sizes: predicting, for example, that a door must be taller than a truck even in a rendering wherein the truck has been scaled down. A further failure attends unusual visualizations, toy miniatures or architectural models among them, wherein familiar proportions are deliberately inverted; in these cases models cleave to their priors and disregard the actual geometric evidence visible within the image.

★ **Analysis.** These behaviors indicate that models lean heavily upon the distributional statistics of object categories rather than performing genuine geometric inference. In lieu of reasoning about projection geometry, relative bounding-box sizes, or depth cues, they substitute their world knowledge of typical scales. This shortcut serves well in everyday contexts yet fails catastrophically in edge cases that deviate from the training distribution. When an unusually small airplane is rendered beside a person, for instance, the model persists in predicting the airplane to be larger. Such errors throw into relief the want of grounding in pixel-level metric cues together with the absence of consistent recourse to perspective or vanishing points.

⊗ **Root cause.** Current architectures incorporate no explicit geometric reasoning modules capable of preserving projection-aware relationships. Supervision for scale and fit tasks is scant, and so models learn correlations between object-category labels and size in place of the underlying geometry. Training objectives, moreover, privilege overall accuracy over fine-grained metric consistency, whence scale-sensitive errors are under-penalized. Bereft of inductive biases for 3D reasoning or of targeted supervision, the model defaults to memorized priors concerning canonical sizes rather than interpreting the actual perspective geometry present within the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** The determination of which object stands in front, behind, or partially concealed remains a core weakness. Models frequently misreport the foreground object where two animals overlap, or where semi-transparent barriers such as glass or fences intervene. In an image in which a cat sits partially behind a curtain, for instance, models may erroneously describe the curtain as occluded by the cat. Thin or fragile structures, wires, tree branches, or railings among them, prove particularly error-prone, being frequently omitted altogether and thereby engendering incorrect answers to questions such as “what object lies behind the fence.” These mistakes recur consistently, most conspicuously where occlusion cues are subtle or where the depth difference is small.

★ **Analysis.** Such failures indicate that models treat depth and occlusion as weak, secondary signals. Although the pixel data harbor unambiguous occlusion cues (e.g., edge boundaries and T-junctions), the intermediate features the model learns do not retain them at sufficient resolution. Aggregation layers blur boundaries and downweight thin structures, in effect expunging the very cues necessary to resolve ordering. It is for this reason that models succeed at gross depth distinctions (foreground versus background) yet collapse in fine-grained cases wherein partial occlusion must be inferred; in practice they guess rather than infer, exhibiting systematic biases toward treating the visually larger or brighter object as the foreground.

⊗ **Root cause.** The architecture wants explicit mechanisms for multi-scale feature retention and provides no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are readily lost once features are pooled across large receptive fields. The training data, furthermore, frequently underrepresent subtle occlusion cases, especially those implicating transparency or fine boundaries. The learned representations consequently flatten layered scenes into 2D appearance maps, thereby precluding the robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models at times conflate perceptual attributes (such as color, texture, or material) with functional or semantic roles. A paradigmatic example is the misjudgment of whether a crosswalk is safe: models may ground the answer solely in the presence of painted stripes while disregarding the state of the traffic light or the approach of vehicles. In like manner, a surface bearing a metallic texture may be misclassified as “steel” even where it is in fact painted plastic. Within tool-use contexts, the presence of handles or familiar shapes may lead the model to impute a functionality that the object does not in fact possess. Such confusions recur frequently on tasks that require the fine-grained visual grounding of attributes.

★ **Analysis.** These errors reveal that language priors strongly bias the model’s predictions wherever the visual input is ambiguous. The decoder favors plausible narratives grounded in co-occurrence within the training data, “crosswalks are safe when stripes are visible,” “shiny textures correspond to metal,” and so forth. This shortcut permits models to produce semantically coherent answers even where they do not inspect the pertinent visual evidence, whence superficially correct reasoning obtains in familiar contexts alongside systematic errors in atypical or adversarial scenarios. In practice, this manifests as overfitting to textual stereotypes in preference to grounding in perceptual detail.

⊗ **Root cause.** The underlying training objective, the maximization of the likelihood of fluent and plausible text, does not enforce visual grounding. Provided that the generated answer is semantically reasonable, the loss is minimized, even where the reasoning is incorrect. The model is consequently incentivized to substitute common-sense priors for genuine perceptual inference. In the absence of auxiliary losses or verification steps that bind textual claims to image regions, models will continue to interchange perceptual and semantic attributes.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** For explanatory or why-style questions, models frequently generate coherent yet unsupported narratives. Asked “why is the object leaning,” for example, a model might answer “because the surface is slippery” even though slipperiness cannot be inferred visually. In another instance, a tilted shelf might be explained as “due to an earthquake,” a claim wholly beyond the evidence present within the image. These errors throw into relief a propensity to fill gaps with imaginative but ungrounded rationales, and thereby to produce fluent explanations that mislead users as to the true evidence.

★ **Analysis.** The issue issues from a decoupling of text generation from visual grounding. The model

generates high-quality language that reads convincingly, and yet no mechanism ensures that these rationales are bound to actual visual features. Confidence calibration compounds the problem, in that models frequently assign high certainty to fabricated explanations, thereby conveying a false impression of reliability. Open-ended outputs may consequently appear authoritative while being factually baseless, a mode of failure of particular concern within safety-critical domains, wherein plausible but false rationales could misdirect decision-making.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence rather than factual grounding. Decoders are optimized to produce the most probable continuation of text given the prompt, unconstrained by any requirement linking the explanation back to specific image evidence. The absence of an explicit verification layer or grounding requirement entails that models default to linguistic plausibility over perceptual truth. In the absence of additional architectural or training constraints, open-ended rationalizations will continue to privilege surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

Upon comparing failures across different models, several consistent patterns emerge, and these afford insight into the underlying mechanisms of error. One salient trend is that the disagreements between models are systematic rather than random: certain models consistently rely upon geometric cues such as relative size, depth ordering, and occlusion relationships, whereas others retreat to linguistic priors or common-sense assumptions derived from training text. This engenders predictable divergences, as when two models return different answers concerning which object is “in front” despite viewing the same scene. Errors likewise spike wherever multiple cues must be integrated simultaneously. Tasks that require reasoning jointly about occlusion, relative scale, and perspective prove particularly exacting; apprehending whether a larger object partially occluding a smaller one lies in front or behind, for instance, frequently precipitates consistent mispredictions. Calibration constitutes a further conspicuous issue, in that high-confidence predictions frequently correspond to incorrect answers, most notably within open-ended explanations or complex spatial queries. These patterns indicate that errors arise from the manner in which models balance perceptual grounding against linguistic plausibility. Different architectures exhibit distinct tendencies, some overrelying upon visual geometry and others overfitting to language correlations, whence failures constitute not merely isolated mistakes but reflect deeper trade-offs in reasoning strategy. This further implies that any attempt to improve performance ought to accommodate both sources of bias concurrently rather than treating errors as independent events.

I.4 MITIGATION STRATEGIES

In light of these error patterns, several targeted strategies may be pursued to reduce failures without necessitating a wholesale architectural redesign. Geometry-aware supervision, for instance, may be incorporated through the addition of auxiliary tasks such as depth prediction, occlusion ordering, or overlap estimation, which encourage models to respect spatial constraints and thereby improve consistency across predictions. Multi-scale feature retention constitutes a further approach: by recourse to edge-aware pooling or attention mechanisms, models may retain information from thin or partially occluded structures that would otherwise vanish under standard downsampling. Coordinate anchoring can help to resolve confusion between viewer-centric and object-centric frames by representing reference points for relative positioning explicitly. Counterfactual augmentation may also be applied, whereby distractor objects are systematically swapped or perturbed so as to compel the model to rely upon the true referent cues in preference to heuristics. Lightweight verification modules can, moreover, be appended atop the existing outputs, cross-checking claims against visual or geometric constraints so as to filter implausible predictions. Each of these strategies admits of incremental implementation, and thereby furnishes clear avenues for reducing systematic failures while leaving the core model largely unaltered.

In addition to these architectural and training augmentations, supervised fine-tuning (SFT) offers a practical pathway toward instilling more reliable spatial reasoning. By curating datasets that explicitly foreground failure cases, near-threshold scale comparisons, occlusion ambiguities, or reference-resolution traps among them, SFT can expose models directly to the edge scenarios wherein they would otherwise default to heuristics. Carefully designed instruction-style fine-tuning further encourages step-by-step reasoning, prompting the model to verbalize intermediate spatial relations

rather than leaping to a guess; this not only fortifies grounding but also improves interpretability, inasmuch as errors become traceable to specific reasoning steps. SFT may, moreover, be paired with counterfactual or adversarial augmentations, thereby ensuring that the model does not simply memorize but instead acquires invariances to distractors and context shifts. Significantly, these interventions require no wholesale changes to the architecture: they adapt existing models by reshaping their inductive biases through supervision. In this respect, SFT serves as a lightweight yet powerful complement to geometry-aware and verification-based strategies.

In the present work we additionally performed SFT using 40% of the training data and observed manifest performance gains, most notably on multiple-choice questions. This outcome intimates that targeted supervision assists models in internalizing spatial cues the more effectively and in resolving distractor confusions wherever the answer options are well defined. The improvements on open-ended responses, however, proved more circumscribed. The model persists in producing fluent yet weakly grounded explanations, an indication that SFT alone cannot fully redress rationalization in the absence of verification. These findings reinforce the need to combine SFT with geometry-aware supervision, counterfactual augmentations, and lightweight verifiers in order to attain balanced progress across both structured and free-form tasks.

Our analysis demonstrates that failures are not arbitrary but structured and interpretable, emerging from recurring patterns rather than from random noise. This regularity indicates that VLM errors follow systematic trajectories bound up with the inductive biases of their architectures and training regimes. Geometric reasoning, for example, frequently collapses where occlusion cues are weak, and in such cases language priors assume control, yielding answers that sound plausible yet want visual grounding. In like manner, scale inference drifts where canonical object-size priors dominate over pixel-level metric cues, thereby engendering consistent misjudgments in near-threshold scenarios. These findings accentuate that the gap is one not simply of capacity but of representation: current models want explicit mechanisms with which to anchor visual features to grounded geometric relations.

By adopting the proposed verification protocols together with the targeted mitigation strategies, researchers can address these weaknesses directly. Geometry-aware supervision and counterfactual augmentation would help models to resist shortcuts, while multi-scale feature retention ensures that thin structures and partial occlusions are not discarded. SFT, when judiciously applied, can further fortify structured reasoning by aligning model predictions with task constraints, though it must be conjoined with grounding objectives if overfitting to discrete-choice formats is to be averted. Significantly, verification modules furnish a lightweight means of enforcing consistency between claimed relations and image evidence, thereby counteracting the propensity to rationalize without checking. Considered in aggregate, these steps chart a practical pathway toward systems that reason more nearly as humans do: robust to clutter, interpretable in their failures, and capable of explaining not merely what they predict but why.

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Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

1. Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
2. Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
3. Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
4. Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

1. Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
2. Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
3. Example Output:
Obj1: Small shaded circle (leftmost)
Obj2: Large unshaded square (center)
Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

1. Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
2. Output Format:
JSON array, e.g.:
[
 {{"ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position": "Leftmost"}},
 {{"ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position": "Center"}}
]

Figure 10: The prompts employed for SPATIOXOLVER (Part 1).

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Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: The prompts employed for SPATIOXOLVER (Part 2).

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Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{
  "objects": [...],
  "relations": [...],
  "groups": [...],
  "symmetries": [...],
  "transformations": [...]
}
```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: The prompts employed for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

We present representative benchmark samples for all 30 sub-categories in SPATIALAB, so as to illustrate the diverse spatial reasoning challenges the benchmark encompasses. These samples foreground the variety of visual phenomena at issue, 3D geometry, occlusion, orientation, relational positioning, size and scale, and navigation tasks among them. Considered together, they afford a concrete view of the modes of reasoning that each model must perform if it is to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Some examples from the benchmark (3D Geometry: Gravity Effects) and evaluation results.

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3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- **Gemini-2.5-Flash:** The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- **InternVL3.5-72B:** The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Some examples from the benchmark (3D Geometry: Spatial Containment) and evaluation results.

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3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Some examples from the benchmark (3D Geometry: Volume Comparison) and evaluation results.

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3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- **GPT-5-mini:** Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- **Gemini-2.5-Flash:** The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- **InternVL3.5-72B:** From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Some examples from the benchmark (Spatial Navigation: Accessibility Constraints) and evaluation results.

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Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

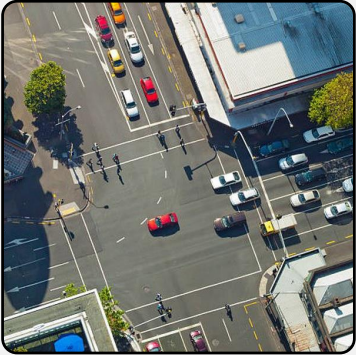
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Some examples from the benchmark (Spatial Navigation: Obstacle Avoidance) and evaluation results.

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Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- Gemini-2.5-Flash: The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- InternVL3.5-72B: The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Some examples from the benchmark (Spatial Navigation: Pathway Existence) and evaluation results.

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Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Some examples from the benchmark (Spatial Navigation: Viewpoint Visibility) and evaluation results.

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Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Some examples from the benchmark (Spatial Navigation: Spatial Sequence) and evaluation results.

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Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Some examples from the benchmark (Size and Scale: Scale Consistency) and evaluation results.

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Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Some examples from the benchmark (Size and Scale: Shadow-Size Projection) and evaluation results.

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Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Some examples from the benchmark (Size and Scale: Perspective Distortion) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- Gemini-2.5-Flash: The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- InternVL3.5-72B: The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- Kimi-VL-A3B-Thinking-2506: Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position. The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- **Gemini-2.5-Flash:** The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- **InternVL3.5-72B:** The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- **Kimi-VL-A3B-Thinking-2506:** The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Some examples from the benchmark (Depth and Occlusion: Complete Occlusion Inference) and evaluation results.

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Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Some examples from the benchmark (Depth and Occlusion: Layering Order) and evaluation results.

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Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Some examples from the benchmark (Depth and Occlusion: Partial Occlusion) and evaluation results.

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Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Some examples from the benchmark (Depth and Occlusion: Reflective Surfaces) and evaluation results.

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Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

- 7
- 6
- 5
- 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Some examples from the benchmark (Depth and Occlusion: Transparency Effects) and evaluation results.

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Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Some examples from the benchmark (Orientation: Cardinal Direction) and evaluation results.

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Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

1. Up
2. Down
3. Away
4. Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- Gemini-2.5-Flash: The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- InternVL3.5-72B: If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- Kimi-VL-A3B-Thinking-2506: The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Some examples from the benchmark (Orientation: Facing Direction) and evaluation results.

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Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

- 90
- 45
- 180
- 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Some examples from the benchmark (Orientation: Object Rotation) and evaluation results.

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Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Some examples from the benchmark (Orientation: Stacking Orientation) and evaluation results.

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Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Some examples from the benchmark (Orientation: Tool Handedness) and evaluation results.

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Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Some examples from the benchmark (Relative Positioning: Alignment Patterns) and evaluation results.

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Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- Gemini-2.5-Flash: The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- InternVL3.5-72B: The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- Kimi-VL-A3B-Thinking-2506: The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Some examples from the benchmark (Relative Positioning: Betweenness Relationships) and evaluation results.

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Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Some examples from the benchmark (Relative Positioning: Corner/Angle Positioning) and evaluation results.

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Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Safari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Some examples from the benchmark (Relative Positioning: Directional Relations) and evaluation results.

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Relative Positioning : Proximity Gradients

Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- **Gemini-2.5-Flash:** "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- **InternVL3.5-72B:** The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- **Kimi-VL-A3B-Thinking-2506:** Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Some examples from the benchmark (Relative Positioning: Proximity Gradients) and evaluation results.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

In order to establish that the benchmark results are not artifacts of random initialization or dataset noise, we evaluate both the robustness of models across runs and the stability of the dataset across repeated trials. Analyses of this kind prove indispensable for small- to medium-scale benchmarks, wherein minor fluctuations are apt to occasion misleading fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010).

Let N denote the number of items *questions* in the benchmark and $R = 3$ the number of independent runs of a given model. Each run employs a distinct random seed while holding all other hyperparameters constant.

K.1 MODEL ROBUSTNESS

In order to interrogate the robustness of model evaluation, we conduct several statistical studies, as described below.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is given by:

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

and its corresponding standard deviation is:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy per run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and its standard deviation across runs is:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ is the overall mean accuracy across all runs.

K.1.2 INTRA-CLASS CORRELATION (ICC)

In order to quantify the agreement between runs, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979) by means of a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

wherein $\sigma_{\text{between}}^2$ denotes the variance between items and σ_{within}^2 the residual variance across runs. ICC values in excess of 0.75 betoken good reliability, while values above 0.9 imply excellent consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

We further analyze the stability of the dataset itself, that is, whether question difficulty remains consistent across runs.

K.2.1 RESAMPLING STUDY

In order to assess **robustness** further, we perform a resampling analysis. From the total of $C = 30$ subcategories, we randomly draw subsets of size $S \in 20, 25$ for each sub-category from each of the $R = 3$ runs, across both models and evaluation types (MCQ and open-ended).

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

The mean accuracy across runs for each subset size is:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

and the corresponding standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary check, the Wilcoxon signed-rank test appraises the median differences between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ denote the difference in accuracies for run r , with ties excluded ($d_r \neq 0$). The test statistic is computed as:

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of signed ranks of positive and negative differences, respectively. The standardized z -score is given by:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

wherein n denotes the number of nonzero pairs.

Should the resulting p -value satisfy $p > 0.05$, we again fail to reject H_0 , thereby confirming that the accuracies obtained from the 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , its difficulty may be defined as $d_i = 1 - \bar{a} * i$. Provided that item difficulty is consistent, the correlation between runs ought to be high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) intimates stable item difficulty and reliable dataset behavior.

K.2.3 CRONBACH'S ALPHA

In order to gauge the overall internal consistency, we compute Cronbach's alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

wherein s_r^2 denotes the variance of run r and s_T^2 the variance of the total scores aggregated across runs. Values of $\alpha \geq 0.9$ betoken excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We perform the analysis on two models, namely one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**, thereby encompassing both large- and small-scale models as well as the proprietary and open-source paradigms. We conduct, in addition, all tests under both the multiple-choice (MCQ) and open-ended evaluation formats (each treated as binary: correct or incorrect). For the statistical analysis, we employed the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for wilcoxon test; `pingouin`³ (Vallat, 2018) for intraclass correlation and cronbach alpha; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We analyzed the stability of model accuracy across $R = 3$ independent runs. The standard deviation of accuracy (σ_A) across runs proves negligible ($< 0.6\%$), an indication that the reported scores constitute precise estimates unperturbed by random-seeding noise.

Table 26: Model accuracy and variance across three independent runs.

Model	Format	Mean Accuracy (A)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The consistently low standard deviations across all models and evaluation formats indicate that the performance estimates are highly stable and reproducible, exhibiting minimal sensitivity to stochastic run-to-run variation. This stability intimates that the observed accuracy differences meaningfully reflect underlying model capability rather than noise introduced by random initialization or decoding. Notably, although open-ended evaluation exhibits marginally higher variance than MCQ, the absolute magnitude remains negligible, a fact that confirms the robust assessment of free-form generation under the current protocol. These findings imply that the dataset size and scoring methodology suffice to yield convergent accuracy estimates without recourse to extensive repeated runs. Consequently, single-run evaluations are in all likelihood reliable for large-scale comparisons, whereas multi-run analyses serve principally to validate robustness. On the whole, this consideration strengthens the empirical credibility of the reported results and underwrites confident cross-model and cross-format interpretation of performance.

Intra-Class Correlation (ICC)

We computed ICC(3,k) in order to quantify the reliability of the scoring mechanism.

Table 27: Intra-Class Correlation (ICC) reliability scores.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The consistently high ICC(3,k) scores demonstrate that the evaluation exhibits near-perfect reliability across independent runs, irrespective of model size or response format. This finding indicates that the item-level ordering of difficulty is preserved and that performance fluctuations leave the relative rankings among samples unaltered. Such strong agreement intimates that the scoring pipeline is

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

internally consistent and insensitive to stochastic variation in the model outputs. Significantly, this consideration precludes random guessing as a dominant factor, even in open-ended generation, wherein variability is typically higher. From an evaluative standpoint, these results imply that the benchmark furnishes a dependable signal for the assessment of model capability, whence longitudinal comparisons and fine-grained ablations may be conducted with high confidence in the stability of measurement.

On the whole, the combined evidence issuing from the multiple-run variance analysis and the ICC reliability demonstrates that the evaluation is highly robust to stochastic effects. Both absolute performance (a low standard deviation) and relative performance (a near-perfect ICC) remain stable across independent runs, a stability that confirms that the results are not driven by random seeding or decoding noise. This robustness ensures that the observed performance gaps across models and formats reflect genuine differences in capability, and the evaluation framework accordingly furnishes a reliable and reproducible basis for comparative analysis and benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study In order to ensure statistical reliability, this resampling procedure is repeated over **1,000 independent trials**, on each occasion drawing fresh random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repetition of the test at this scale stabilizes the distribution of accuracy differences and p -values, and thereby ensures that the conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials accordingly furnishes strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling study (1,000 trials): Wilcoxon signed-rank test ($S = 20$ versus $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-v1-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-v1-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results furnish strong empirical evidence that the accuracy estimates have already converged at relatively small subcategory sizes. The consistently high mean (p)-values, together with the absence of statistically significant differences across all 1,000 trials, indicate that the performance metrics are insensitive to moderate reductions in sample size. This stability obtains uniformly across models and evaluation formats, an indication that neither model scale nor output structure introduces additional sampling uncertainty. Significantly, these findings imply that the variance in accuracy is dominated by model behavior rather than by data sparsity at the subcategory level. The current subcategory size (≈ 50) may consequently be deemed statistically redundant, affording a substantial margin of safety beyond the minimum required for reliable estimation. This consideration corroborates the validity of the benchmark design and intimates that future extensions might reallocate samples across categories without any compromise of statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We measured the mean pairwise Pearson correlation ($\bar{\rho}$) between runs together with Cronbach’s Alpha in order to assess internal consistency.

Table 29: Item-level consistency metrics.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-v1-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-v1-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The exceedingly high mean pairwise correlations (> 0.94) demonstrate strong item-level stability, an indication that the relative difficulty of individual questions is preserved across independent runs. This consistency evidences that model responses are systematically structured rather than subject to random variation, even under stochastic decoding. The comparably high Cronbach’s α values (> 0.9) further confirm that the dataset exhibits excellent internal coherence, its items contributing reliably

to the overall accuracy measure. Notably, the slight reductions observed in the open-ended formats are to be expected given their greater expressive flexibility, and yet remain well within the range of excellent reliability. Considered together, these results indicate that the benchmark functions as a unified and internally consistent measurement instrument, whence differences in model performance may be confidently ascribed to genuine variation in capability rather than to instability at the item level.

K.3.3 OVERALL DISCUSSION

Our expanded statistical testing establishes that our evaluation and dataset are robust: **ICC and Cronbach’s α scores consistently exceed 0.98**, and the **standard deviation of accuracy across runs is negligible**. The **resampling study**, furthermore, confirms that even smaller subsets yield results statistically indistinguishable from those of the full set, and thereby demonstrates that our subcategory sizes (~ 50) suffice for robust conclusions. These results establish that the granular analysis presented in the paper reflects genuine differences in model capability rather than statistical noise.

Our expanded statistical testing establishes that the evaluation protocol and dataset are highly robust, with **ICC and Cronbach’s α scores consistently exceeding 0.98**, an indication of near-perfect reliability and internal consistency. The **standard deviation of accuracy across runs remains negligible** ($< 0.6\%$), a fact that confirms the stability of the reported model performance and its independence from stochastic effects such as random seeding or decoding variability. The **resampling study**, in addition, demonstrates that even substantially smaller subsets ($N = 20$ and $N = 25$) yield accuracy estimates statistically indistinguishable from those of the full set, and thereby establishes that the chosen subcategory size (~ 25) lies well beyond the threshold requisite for convergence. Collectively, these results show that both the absolute scores and the relative performance rankings are stable across runs and sampling regimes. This statistical robustness ensures that fine-grained comparisons across models, formats, and subcategories are meaningful. Consequently, the granular analyses reported in the paper reflect genuine differences in model capability rather than artifacts of sampling variance or evaluation noise.